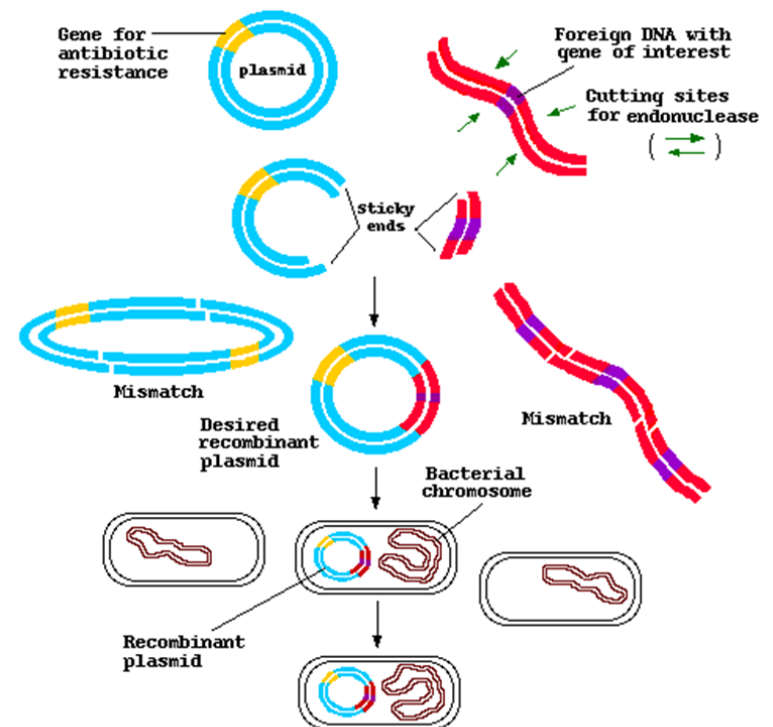


Molecular Cloning

Lecture 10



Plasmid Insertion



Molecular Biology:

Molecular Biology: made simple and fun (3/e)

David P. Clark and Lonnie D. Russell

Presentation prepared by Prof. Woo-Yiel Lee

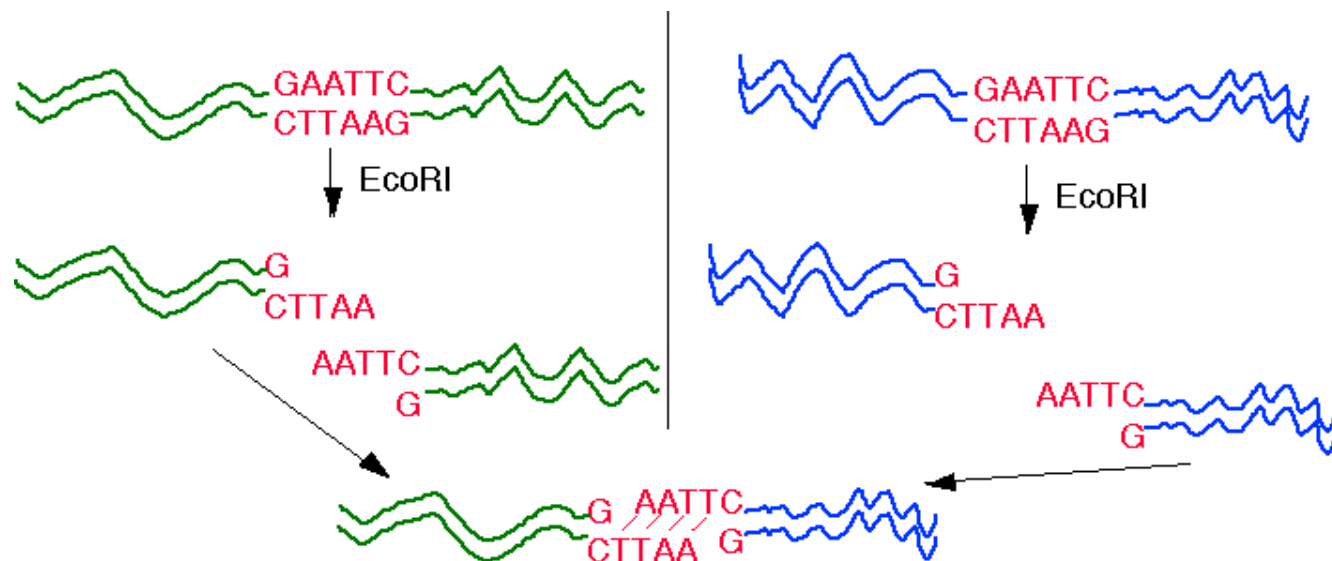
➤ Gene Cloning

- Gene cloning links eukaryotic genes to small bacterial or phage DNAs and inserting these recombinant molecules into bacterial hosts
- One can then produce large quantities of these genes in pure form

Lecture 10_1. 재조합 DNA 제작

1. 재조합 DNA (DNA Recombination)
 - Gene cloning
 - Gene Library
 - PCR
 - Cloning vectors
2. 유전자의 대장균내 도입 방법
 - 형질전환 (transformation)
3. 형질전환된 세포의 선발
 - Screening
 - 재조합 plasmid 확인
4. 재조합 DNA 기술의 전 과정

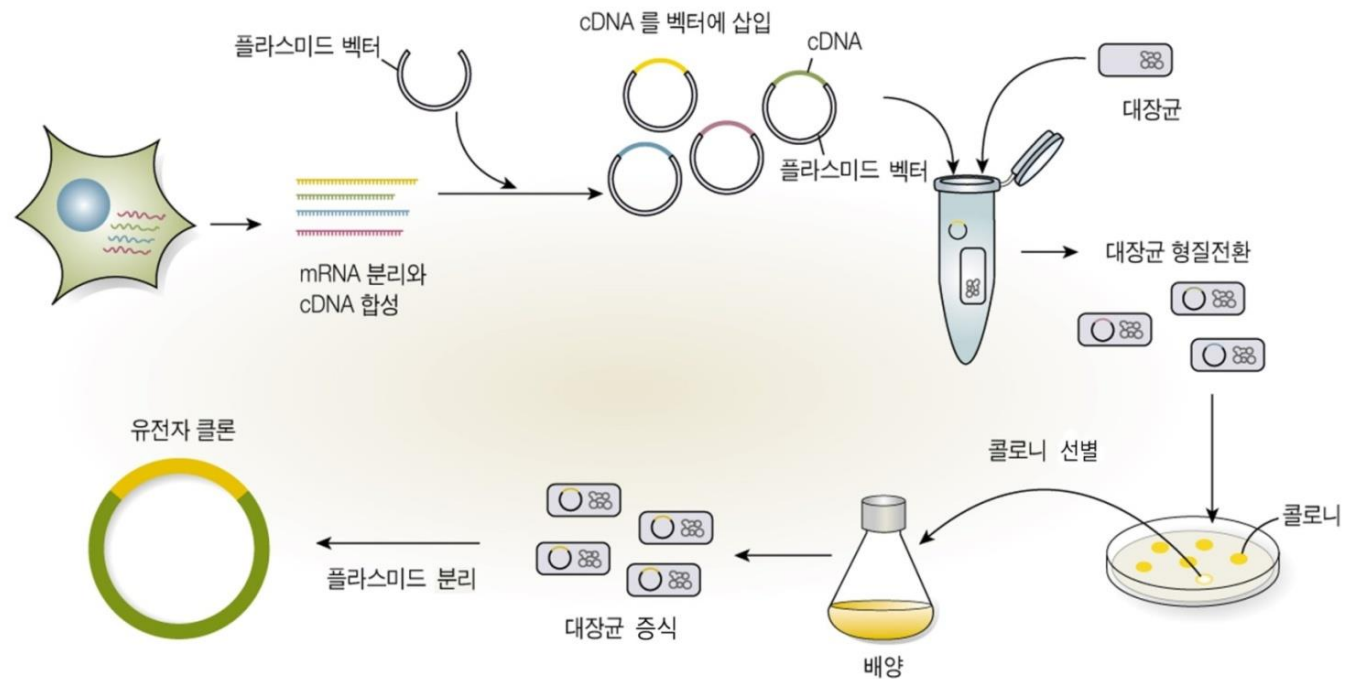
1. DNA Recombination



- 제한효소에 의한 재조합 DNA분자의 제작
- 플라스미드 벡터를 이용한 재조합 DNA 분자의 제작
- 벡터 DNA와 삽입시킬 DNA를 특정 제한효소로 절단

➤ Gene Cloning

- 여러 유전자 집합에서 한 종류의 유전자를 분리하고 증폭하는 과정
- S. Cohen & H. Boyer: 1973년 대장균에 플라스미드를 도입하여 유전자 클로닝에 성공



- **Gene libraries:** 한 생명체의 조직이나 세포의 모든 유전자를 포함하는 DNA 클론 집합
 - Genomic library
 - cDNA library
 - Library 제조 후 여러가지 screening 방법을 이용 target DNA 클론 선별

GENE LIBRARIES

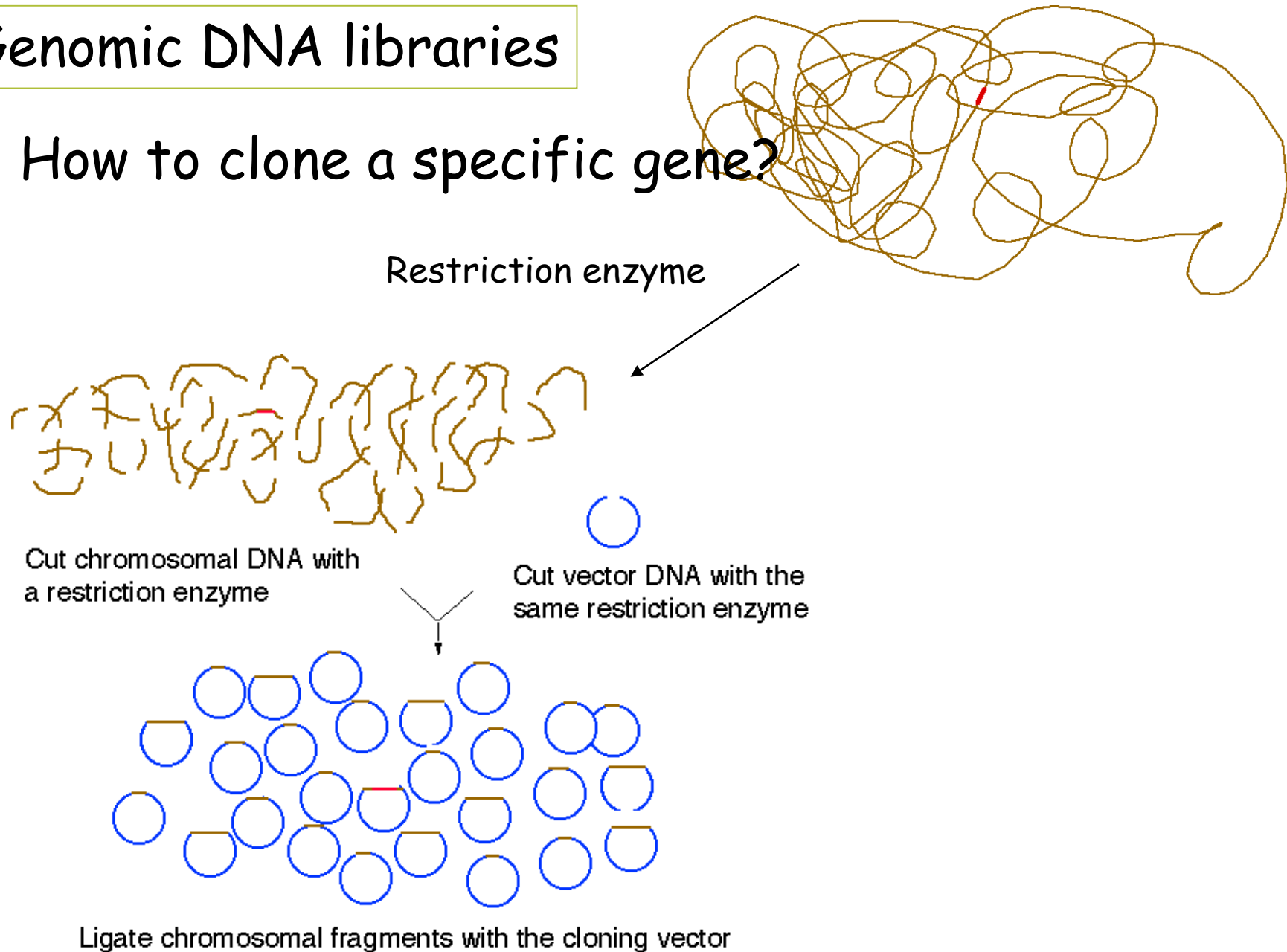
How to clone one particular gene from thousands of genes present in the genome?

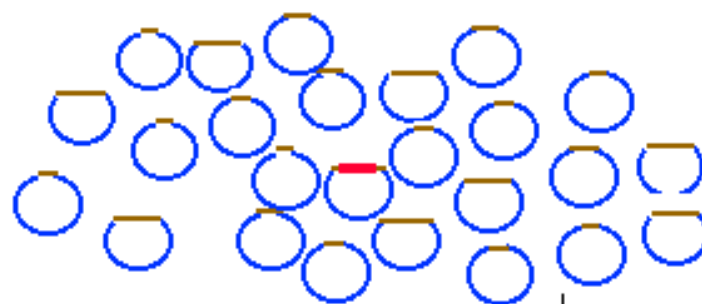
One of the most frequently used way to clone a specific gene is to use a **GENE LIBRARY** approach:

Clone everything you can and then find what you need.

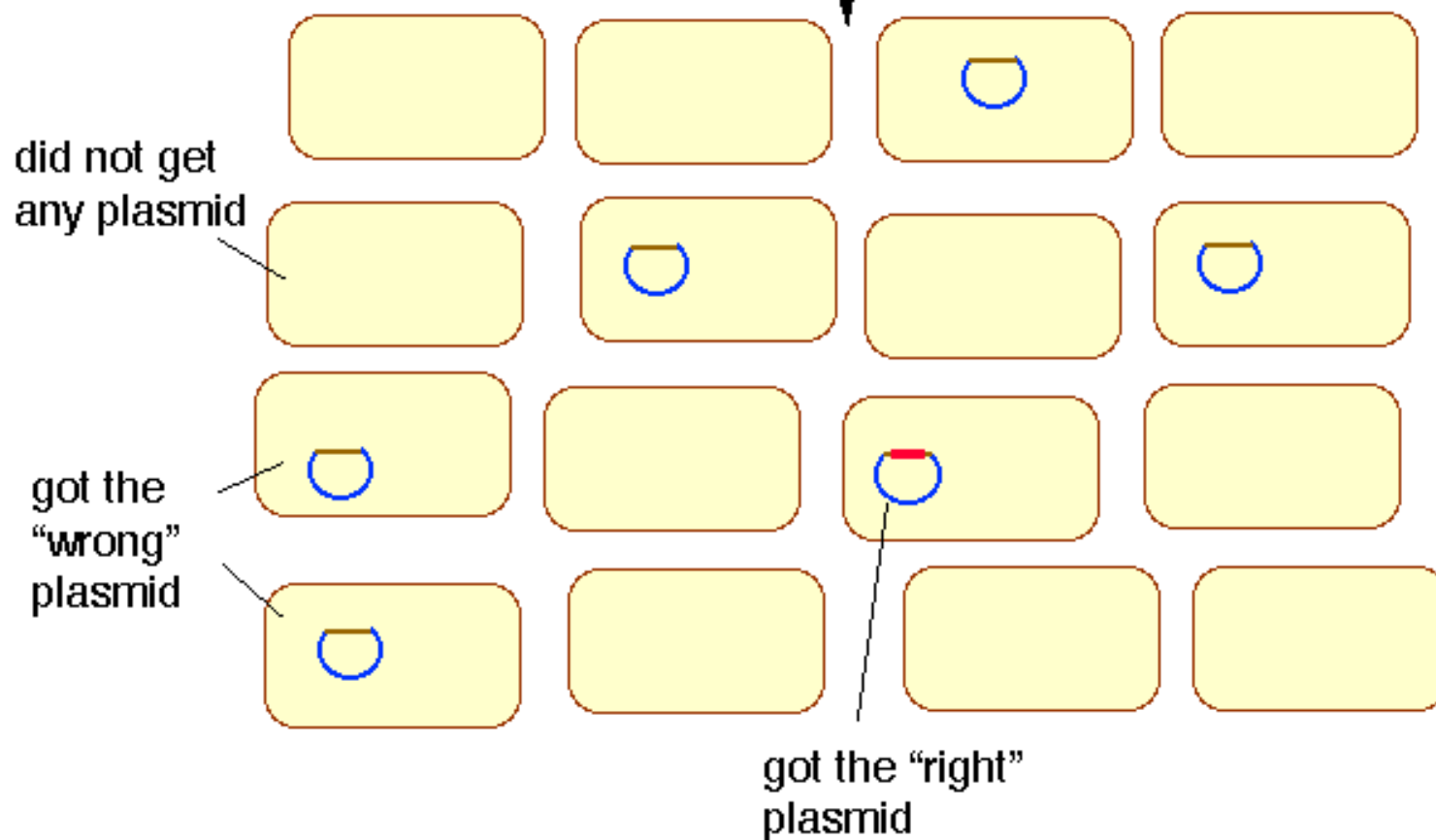
Genomic DNA libraries

I. How to clone a specific gene?

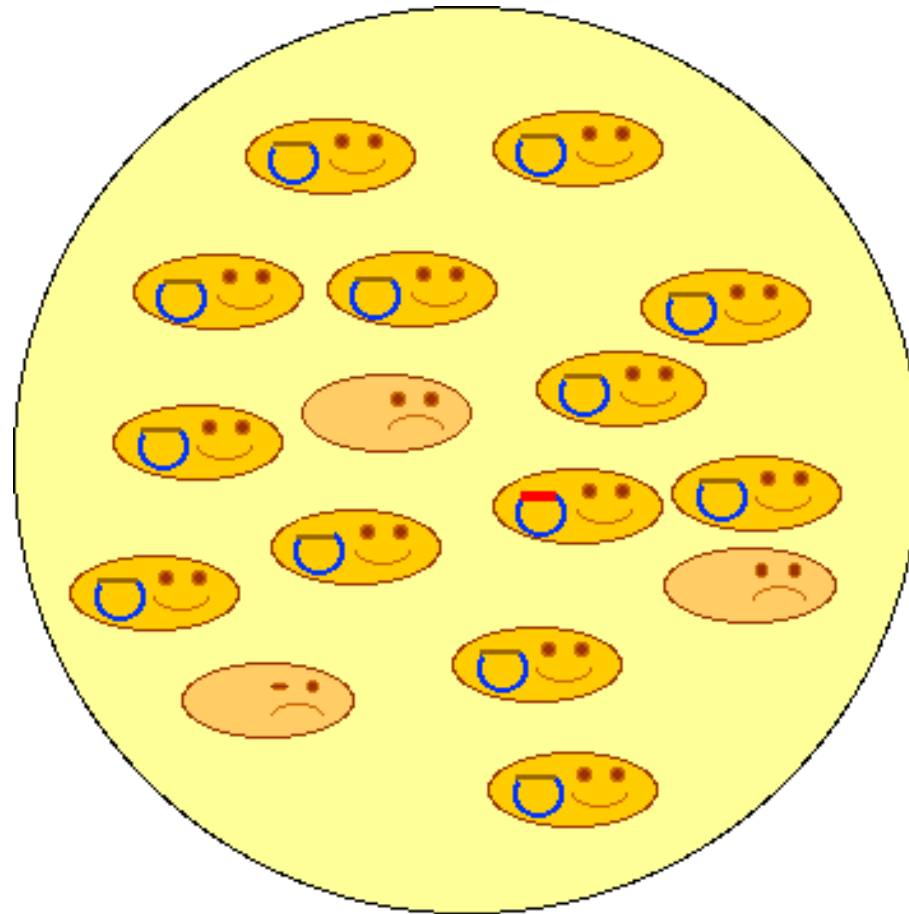




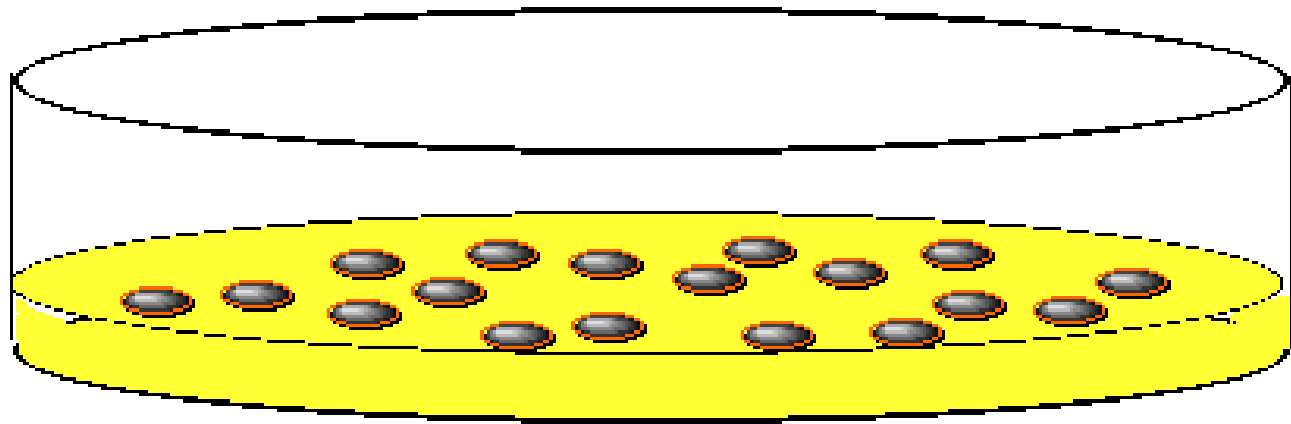
Transform a mixture of recombinant plasmids into competent cells



After plating on agar plates containing the selective antibiotic, only cells with plasmids will survive. Plating makes possible to screen out cells that did not get any plasmid.



Each transformed cell that acquired a plasmid forms a colony on selective medium. All cells in a colony have the same plasmid. Cells in different colonies have different plasmids.



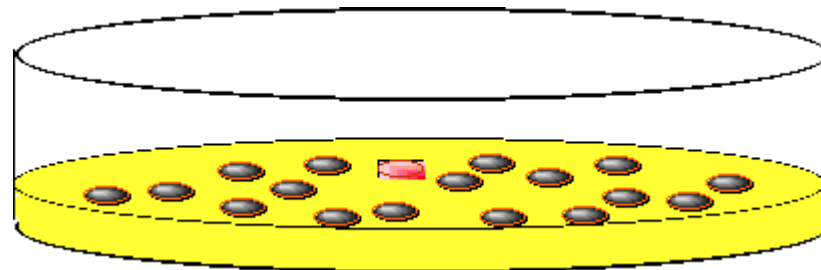
agar medium with antibiotic

II. How to find the clone with "our" gene?

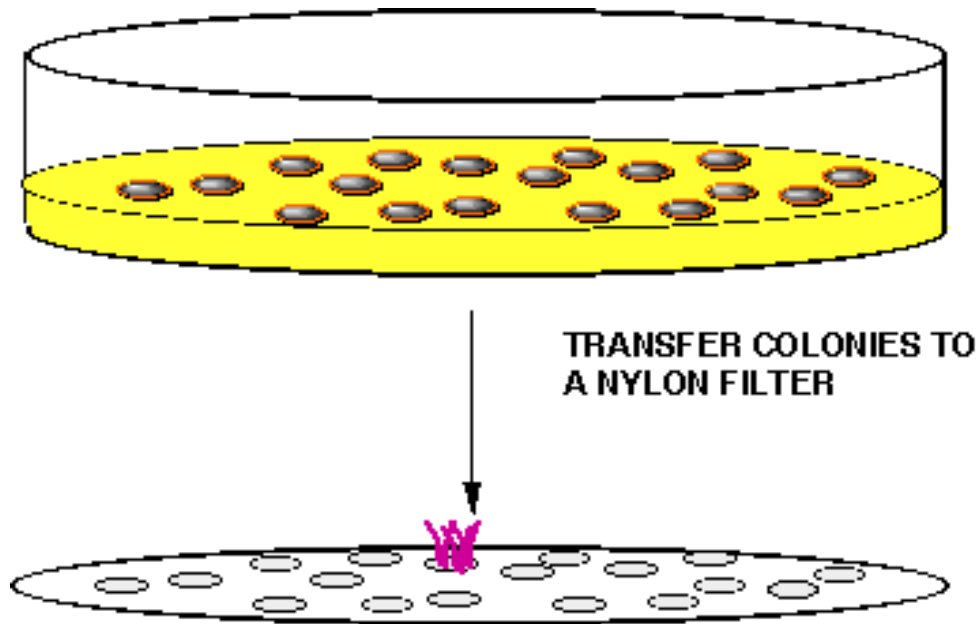
The most common methods include:

- Phenotypic screening
- Screening with antibodies
- DNA hybridization

1. Phenotypic screening is used when cloned gene is expressed and changes properties of the cell in an "obvious way". In the shown example, the protein encoded in the cloned gene changes the color of transformed cells.



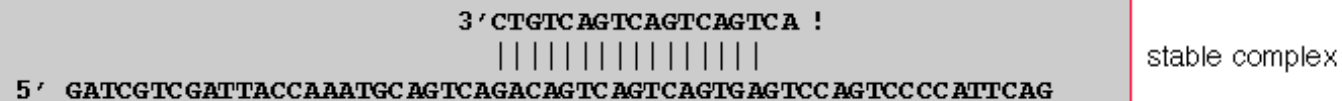
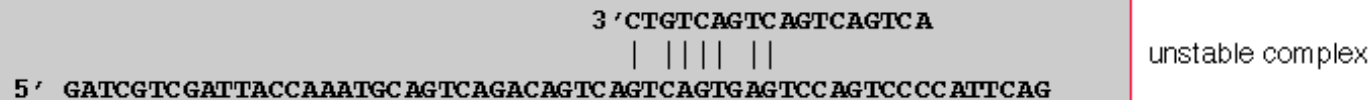
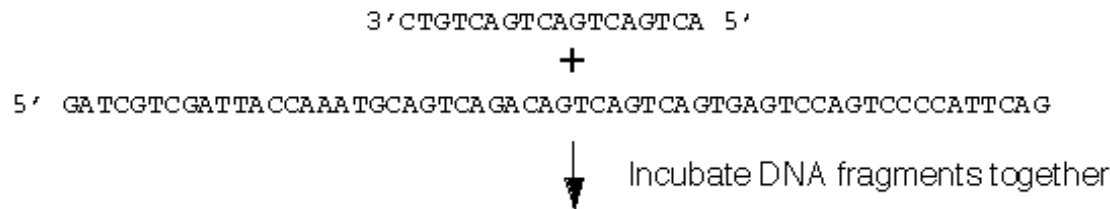
2. Screening with antibodies is used when cloned gene is expressed and antibodies recognizing the encoded protein are available.



Hybridize filter with **antibodies**
that recognize the protein
encoded in "our" gene

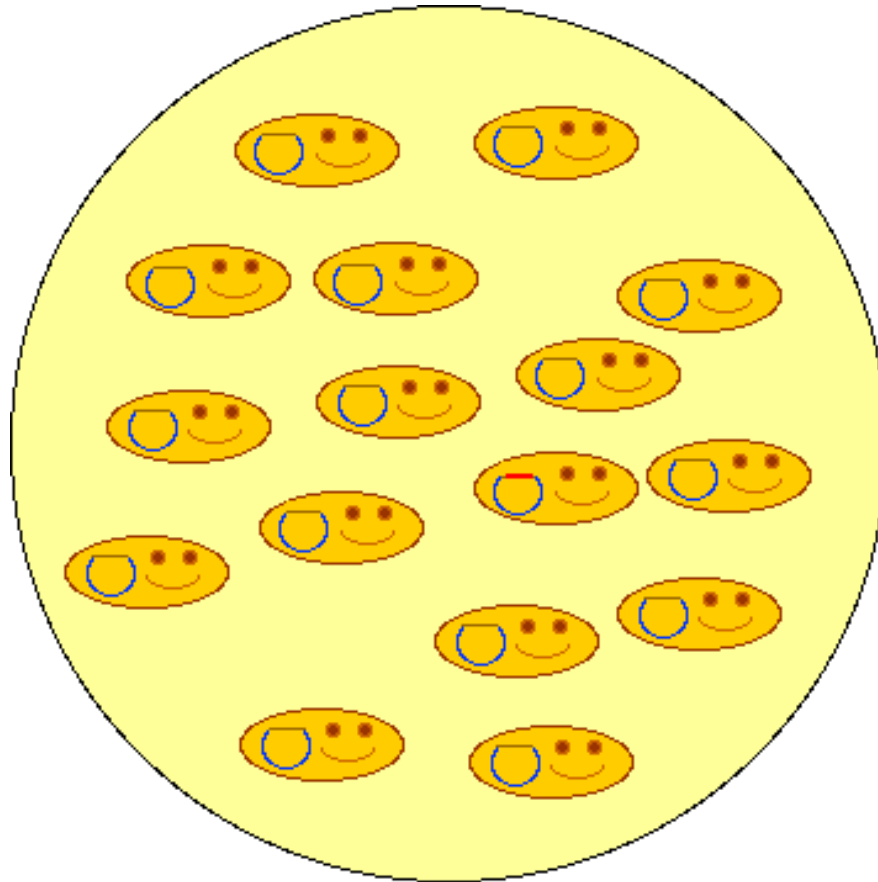
3. DNA hybridization

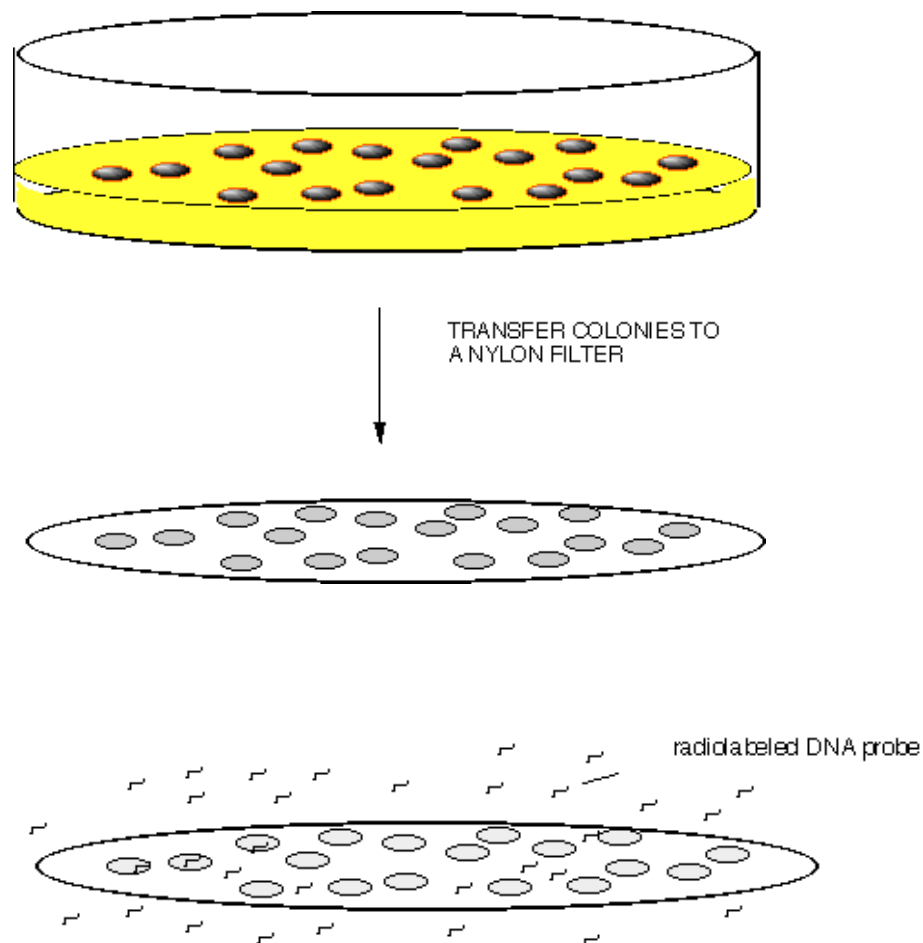
If two single stranded DNA molecules have complementary nucleotide sequences they can **hybridize**: form a stable double stranded complex



If the nucleotide sequence of the whole gene, or any part of it, is known, then DNA hybridization can be used to find the clone in the library that contains the cloned gene.

Only one colony on this plate contains DNA with the nucleotide sequence of the gene we are looking for





1. Synthesize a DNA fragment complementary to any strand of "our" gene.
2. Radiolabel the probe.
3. Lyse cells on the filter and denature DNA.
4. Hybridize the probe with the filter.
5. Wash out excess of the probe.

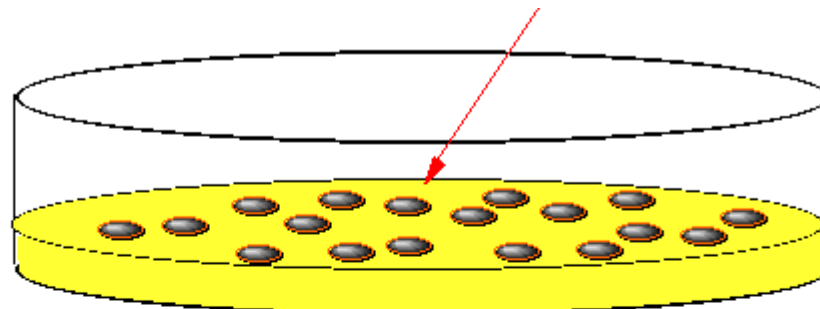


Only the colony that contains plasmids with “our gene” will hybridize with the probe and become radiolabeled.



Expose the filter to the X-ray film: a black spot will appear on the place of a colony with “our gene”.

Master plate



X-ray film



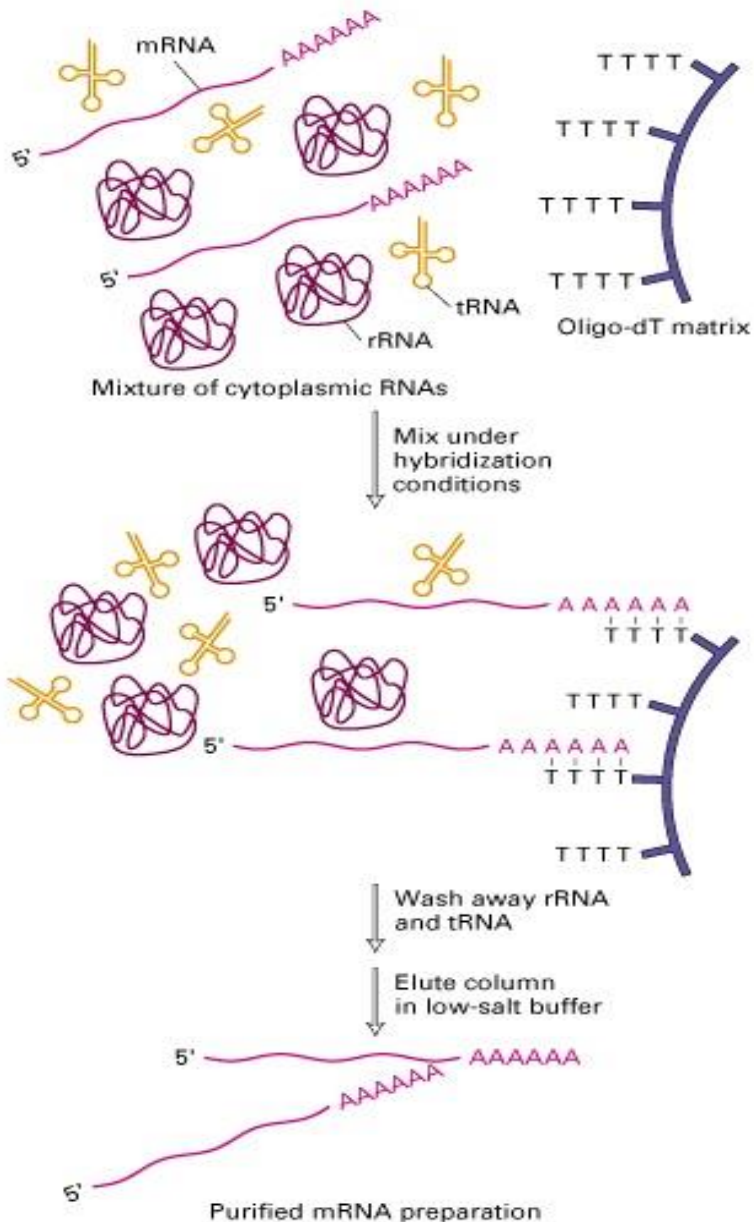
Find the colony on the master plate that contains the cloned gene.

➤ cDNA Libraries 제조

- cDNA: mRNA의 상보적인 DNA
- Reverse transcriptase를 이용 분리한 mRNA로부터 cDNA 합성
 - ✓ Primer가 요구됨
 - ✓ mRNA의 poly A tail를 활용 (oligo dT가 primer로 작용)

cDNA libraries

Isolate mRNA from eukaryotic cells by lysing cells and making use of the fact that at the 3' end of nearly all eukaryotic mRNAs there is a stretch of 50 - 250 adenylate residues the **polyA tail**. This allows them to be purified away from the more abundant tRNAs and rRNAs.



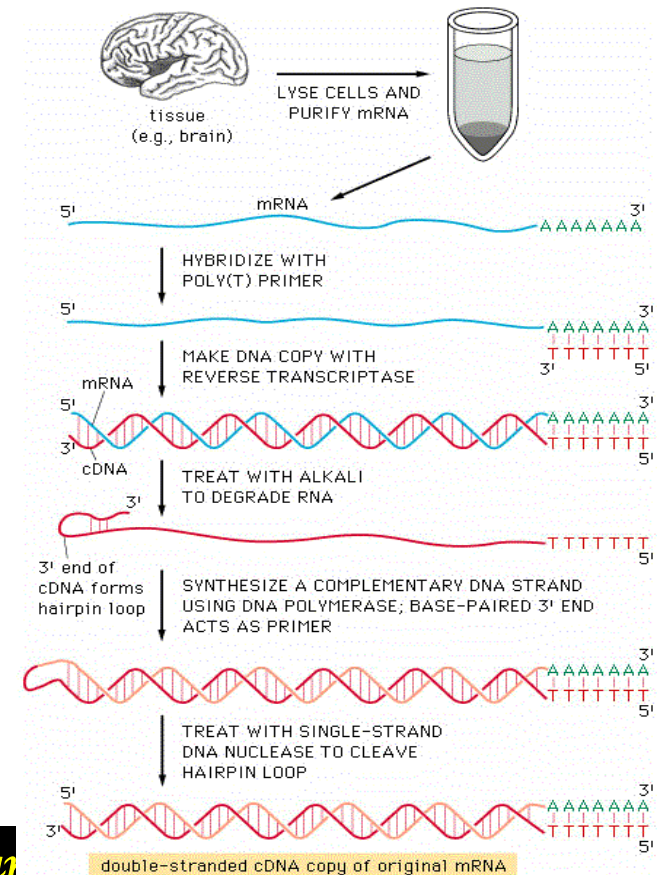
I. Purification

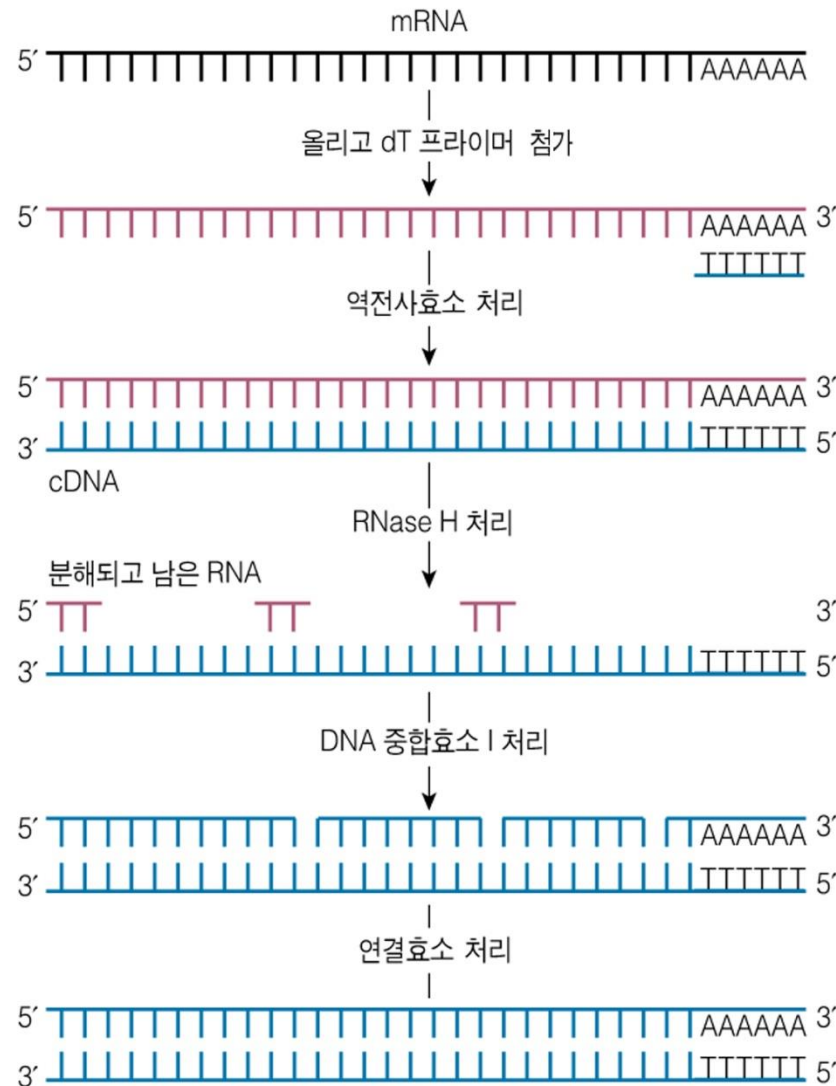
can be performed by binding mRNAs on a solid matrix to which short strings of thymidylate residues are attached (**oligo dT matrix**).

The mRNAs are removed again by washing in a low salt buffer.

II. Preparation of cDNA from mRNA

- mRNA 주형, oligo dT를 primer, reverse transcriptase에 의한 DNA 중합
- Rnase H: mRNA 부분적 제거, 일부 mRNA를 남김
- 남아있는 mRNA를 primer로 활용 **DNA polymerase I**으로 처리
- DNA ligase





III. Preparation of the cDNA for cloning

- Flush cDNA S1 nuclease (remove 3' overhangs)
Klenow enzyme (fill in any missing 3' nucleotides)

(At this stage, can ligate the cDNAs by blunt-end ligation.)

- Ligation of linkers to either end of the cDNAs using **DNA ligase**

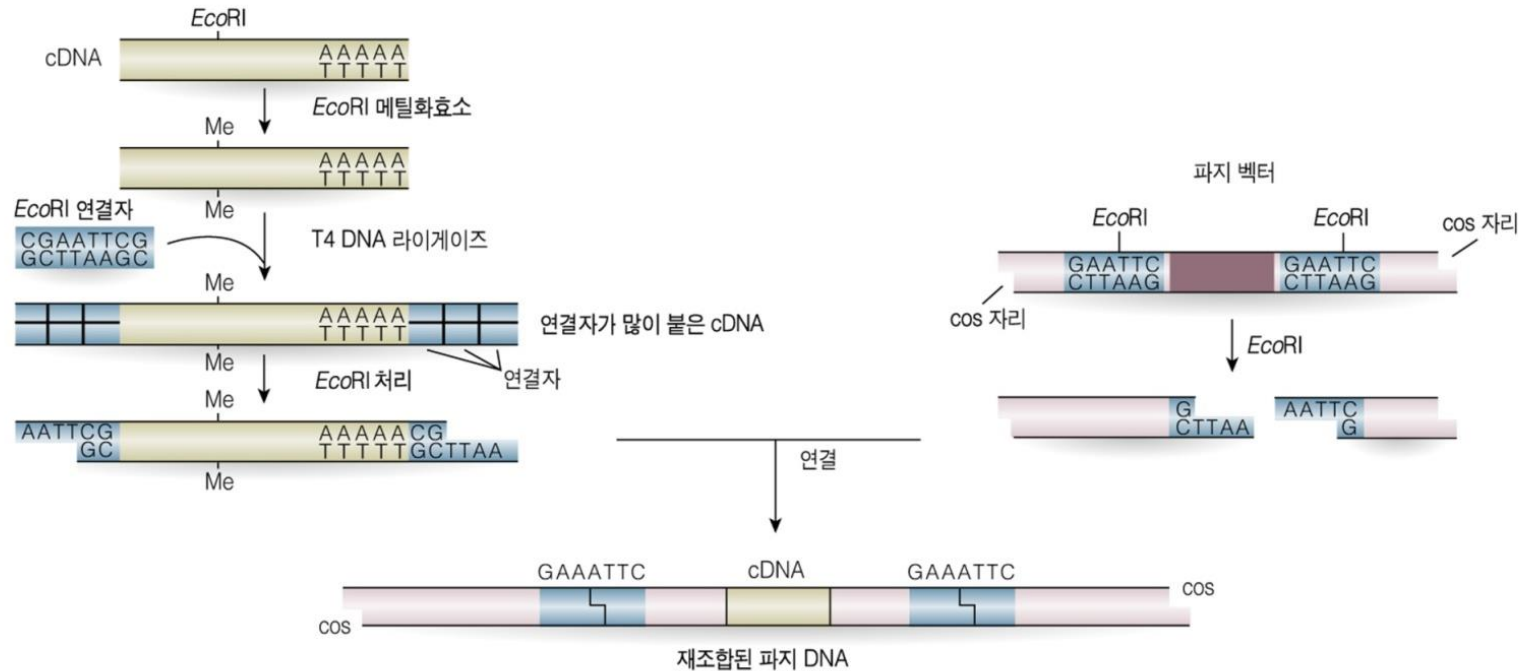
The **linkers** (short oligonucleotides incorporating restriction enzyme sites (e.g. *EcoRI*))

If the 5' ends of the linkers are not phosphorylated, only one can be added to each cDNA end.

- Digestion with *EcoRI*
- Cloning into an *EcoRI*-digested vector.

➤ cDNA 클론 제조 (벡터로 삽입)

- 제한효소자리가 있는 linker를 ligase로 cDNA 말단에 연결



➤ Vectors :

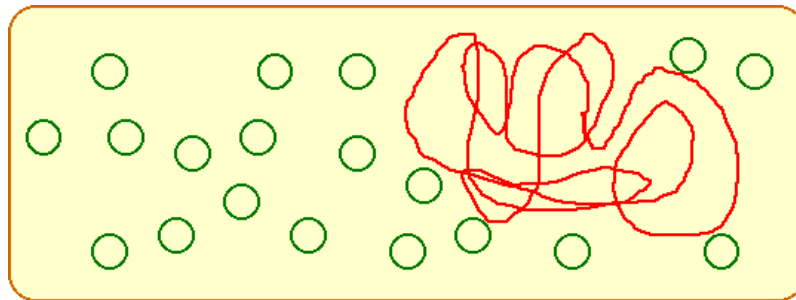
- 외래 DNA와 결합하여 재조합 DNA를 만드는데 쓰이는 DNA vector *유전공학 의미 : DNA 운반체*
- 의학적 의미 : 전달자 = 자신은 직접 병을 일으키지는 않으나 질병을 일으키는 병원균을 한 숙주로부터 다른 숙주로 전달하는 생명을 일컫는 말

In gene cloning, once recombinant DNA is constructed, it is introduced into a bacterial or eukaryotic host. In the host, recombinant DNA has to be maintained, replicated and passed from one generation to another. This is achieved by introducing recombinant DNA into a cell on a **DNA vehicle called cloning vector**. The most commonly used vectors are **plasmid cloning vectors**.

- Bacterial cells may contain extra-chromosomal DNA called **plasmids**. Plasmids are usually represented by small, circular DNA.



- Some plasmids are present in multiple copies in the cell



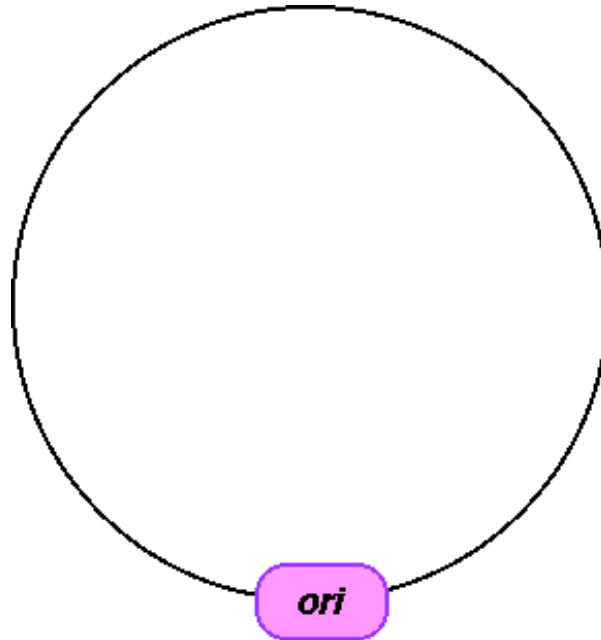
- Artificially constructed plasmids used as cloning vehicles are called **plasmid cloning vectors**
- Three main features of the cloning Vector:

1. ORIGIN OF REPLICATION

2. SELECTABLE MARKER

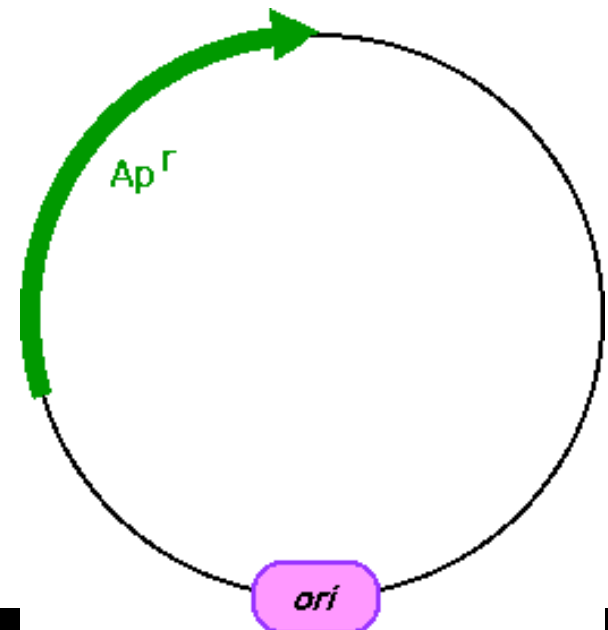
3. A MULTIPLE CLONING SITE

1. Origin of replication is a DNA segment recognized by the cellular DNA-replication enzymes. Without replication origin, DNA cannot be replicated in the cell.



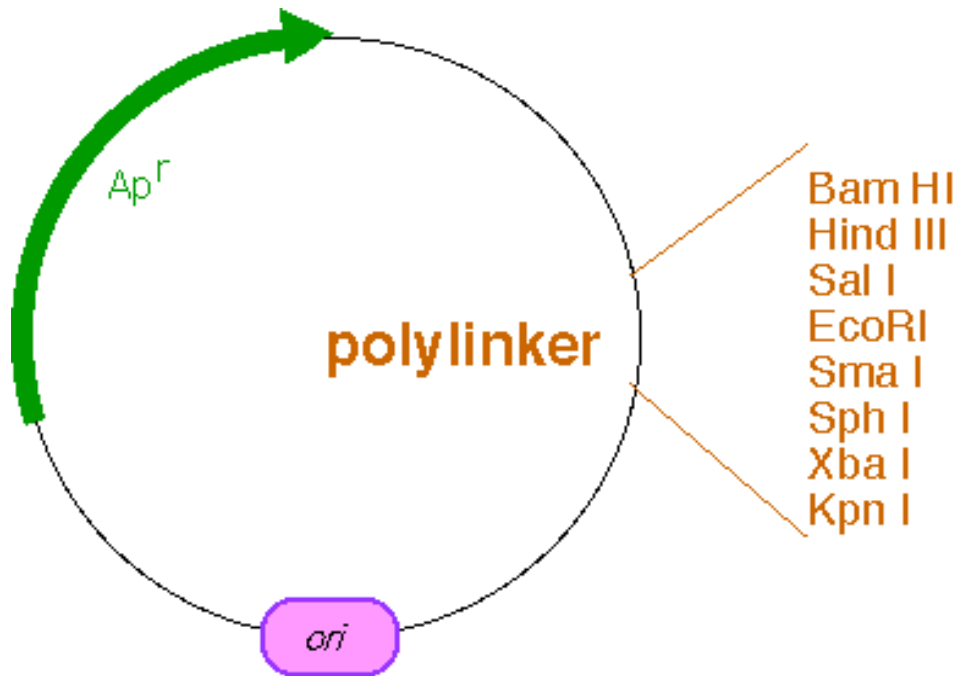
2. Selective marker is required for maintenance of plasmid in the cell.

- Under the selective conditions, only cells that contain plasmids with selectable marker can survive.
- Commonly, genes that confer resistance to various antibiotics are used as selective markers in cloning vectors. (ex: ampicillin, neomycin, or chloramphenicol)



3. Many cloning vectors contain a **multiple cloning site** or **polylinker**:

a DNA segment with several unique sites for restriction nucleases located next to each other



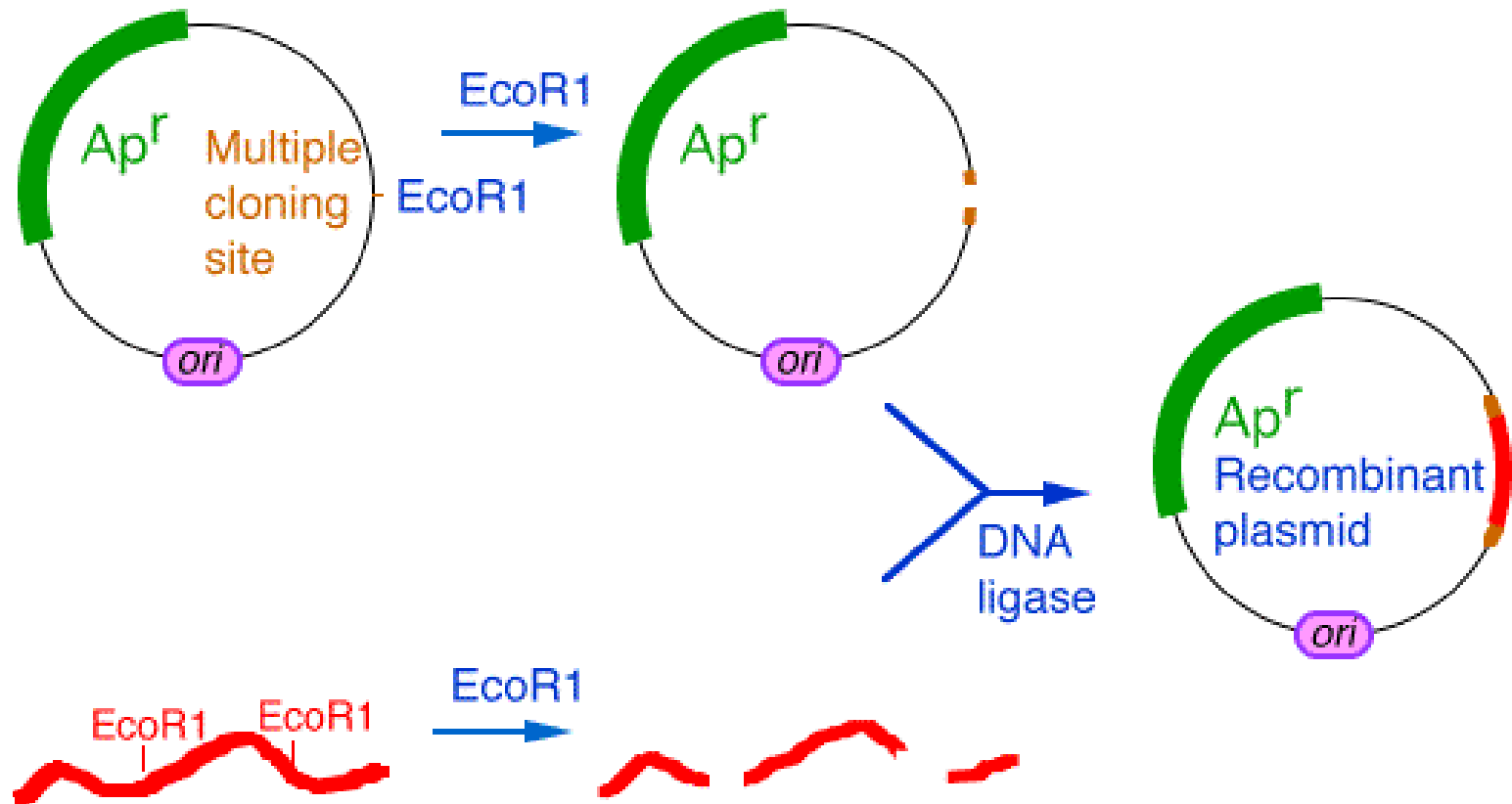
For example, this is a polylinker of one of the commonly used cloning vector plasmids pUC19:

HindIII SphI PstI SalI XbaI BamHI SmaI KpnI SacI EcoRI

5' AAGCTTGCATGCCTGCAGGTCGACTCTAGAGGATCCCGGGTACCGAGCTCGAATTC.3'
3' TTCGAACGTACGGACGTCCAGCTGAGATCTCCTAGGGGCCCATGGCTCGAGCTTAAG.5'

Restriction sites of the polylinker are not present anywhere else in the plasmid. Therefore, cutting plasmids with one of the restriction enzymes that recognize a site in the polylinker does not disrupt any of the essential features of the vector.

Gene to be cloned can be introduced into the cloning vector at one of the restriction sites present in the polylinker

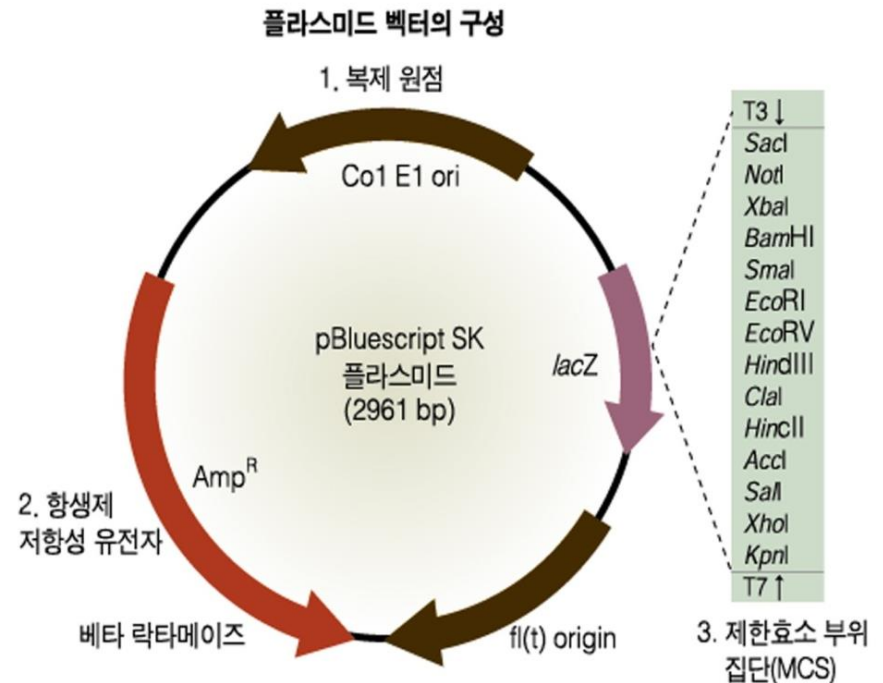


➤ Cloning vector:

- 외부 DNA를 숙주세포로도입한 후 숙주내에서 복제하여 많은 사본 수로 벡터 DNA 자신과 삽입된 외부 DNA를 생산하는 벡터
- 순수DNA를 증식시켜 분리 및 정제를 필요로 할때 사용되는 벡터
 - ✓ 재조합 DNA 제작
 - ✓ 염기서열 결정
- 세균성 플라스미드, 바이러스로부터 얻은 것
- 벡터의 조건
 - ✓ copy number
 - ✓ Simple purification
 - ✓ 외부 유전자 cloning 가능 (multiple cloning site)
 - ✓ Replication origin (ori)
 - ✓ Selection marker (선발표지자)

➤ 플라스미드의 구조와 특징

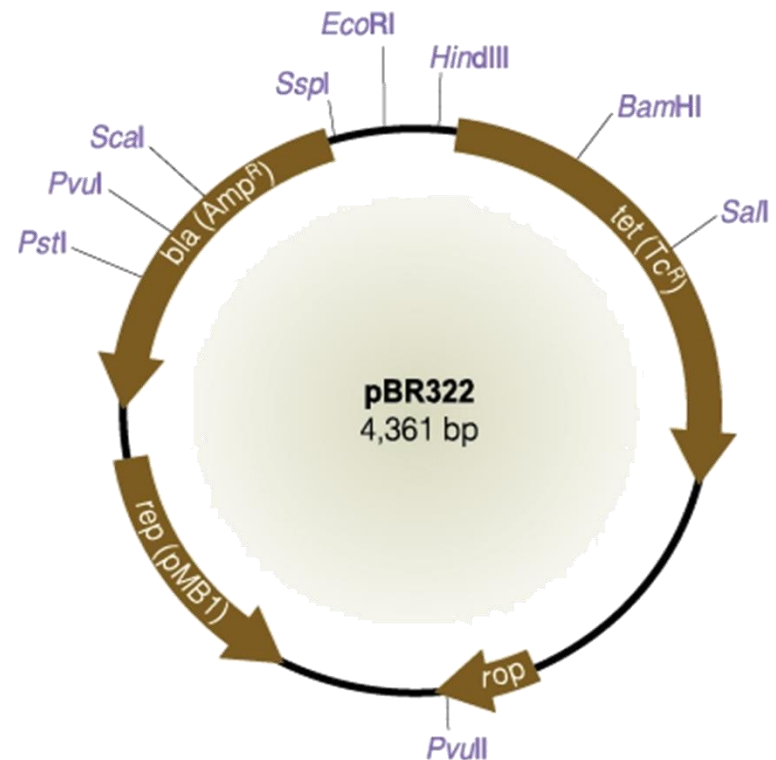
- Replication origin
- Antibiotic resistance gene
- Multiple cloning site (MCS)
- *LacZ* 유전자



➤ 대표적인 플라스미드 벡터의 종류

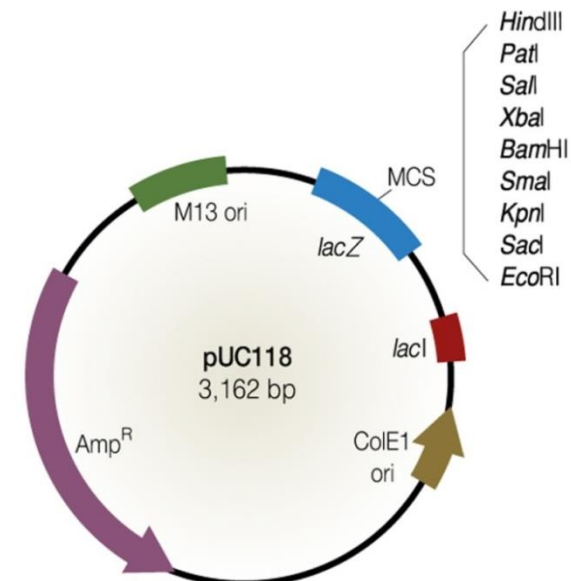
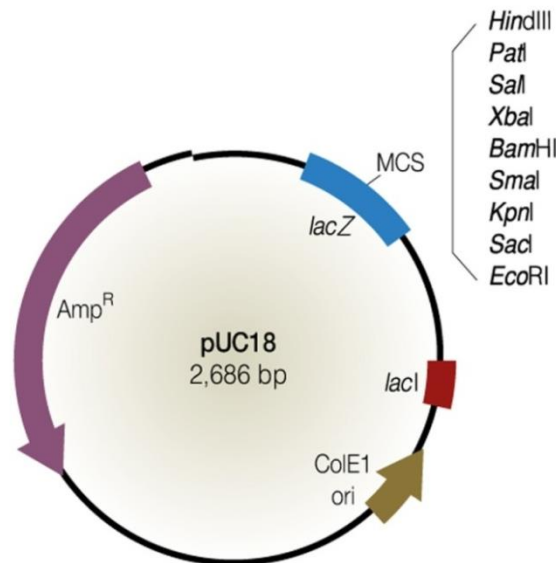
■ pBR322

- ✓ 최초개발
- ✓ 4,361 bp
- ✓ pMB 유래 복제원점인 *rep*
- ✓ 사본 수 조절 *rop* 유전자 (세포당 15-20개)
- ✓ Amp^R , Tet^R 저항유전자



■ pUC, pPU118

- ✓ 2,686 bp
- ✓ Ampicilin 저항성 유전자
- ✓ *LacZ*를 함유 (**white-blue screening**)
- ✓ pBR322에 비한 장점
 - ❑ MCS: 외부 DNA형태를 다양하게 해줌
 - ❑ 500-700/cell 사본수
 - ❑ pUC118, pUC119는 단일가닥 생성을 위해M13파지의 복제원점을 포함

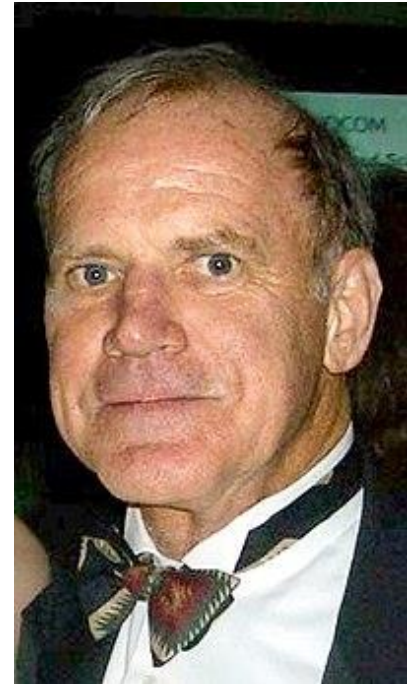


➤ Polymerase Chain Reaction (PCR) (flash)

- Described in 1980's
- Relies on the ability of polymerases to extend oligonucleotide primers (DNA replication) annealed to single stranded DNA
- Discovery and subsequent use of thermostable enzymes had major impact

➤ 캐리 멀리스는 1944년 12월 28일 블루리지 산맥 근처에 있는 노스캐롤라이나주 르느와르에서 태어났다.^[3] 사우스캐롤라이나주 컬럼비아에서 성장하였다.^[4] 1962년 조지아 공과대학교에 입학하여 1966년 학사학위로 졸업하였다

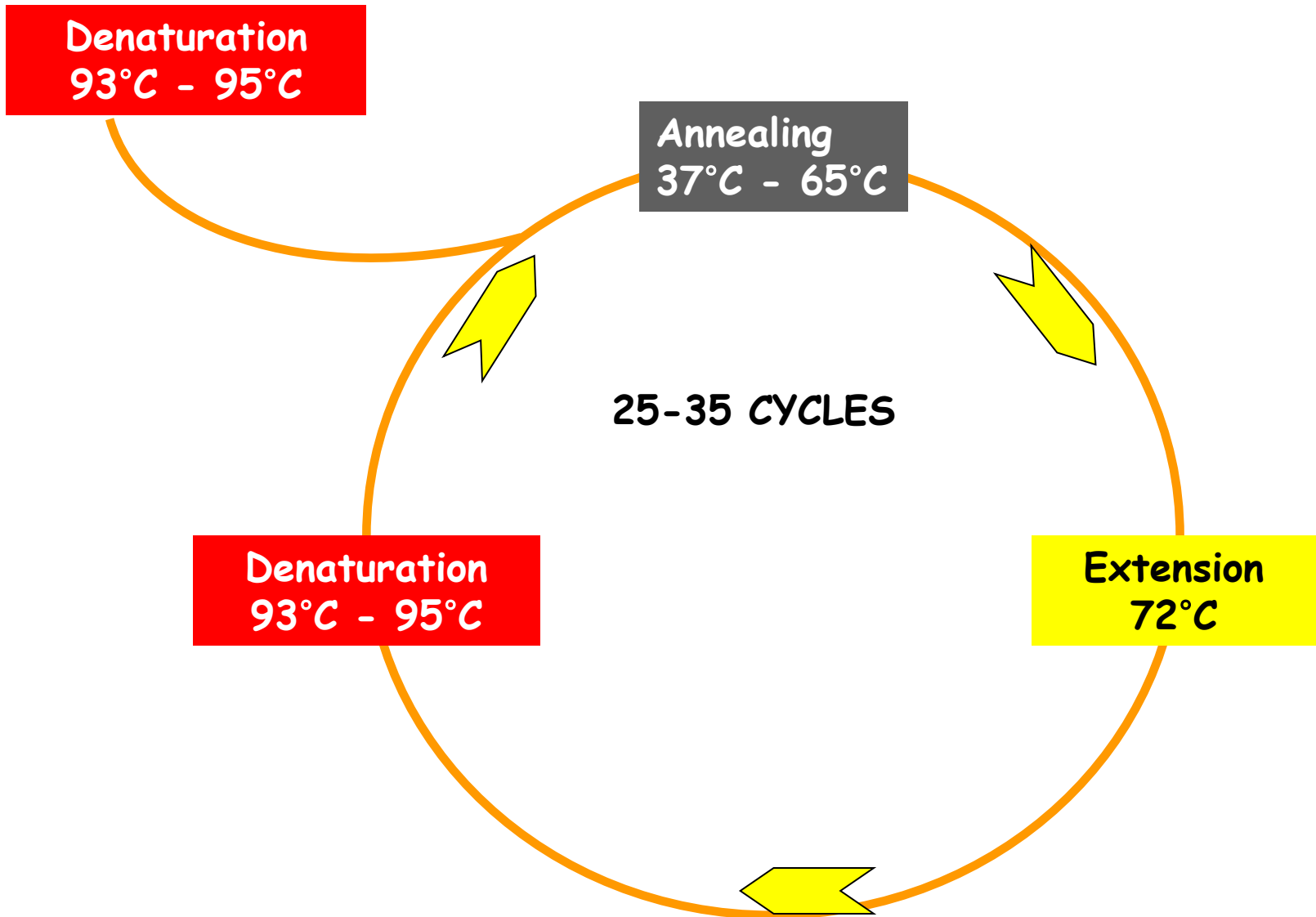
➤ 1972년 캘리포니아 대학교에서 박사 학위를 받았으며 자신의 전공으로 단백질과 효소의 구조 연구를 선택하였다.^[4] 캔자스 대학교에서 2년 동안 근무한 후 캘리포니아 대학교 샌프란시스코의 생화학 교수로 부임하였다.



➤ 1984년에 캐리 멀리스가 특정 DNA 서열을 증폭하기 위한 방법을 고안하여 중합효소 연쇄 반응(Polymerase Chain Reaction)이라고 불렀다.

➤ 1985년에 중합효소를 클레나우 중합효소 (Klenow polymerase)로 사용하는 PCR이 처음 공식적으로 발표됐다. 클레나우 중합효소는 열에 약했기 때문에 매 주기마다 효소를 새로 넣어 주어야 했고 생성물의 최대 길이는 400bp에 불과했다. 1988년에 DNA 중합효소로 *Thermophilus aquaticus*라는 미생물(극 호열균)의 DNA 중합효소인 Taq를 사용한 논문이 발행되었다. 열에 강한 이 중합효소는 PCR의 효율을 월등히 끌어올렸다.

PCR (Polymerase Chain Reaction)



PCR (Step I)

- Template denaturation
 - Breaks H-bonds between complementary bases
 - ✓ 90-95°C
 - ✓ Degradation of even thermostable enzymes
 - Can be used to denature proteases in sample or even to release DNA from sample (e.g. bacteiral colony lift)

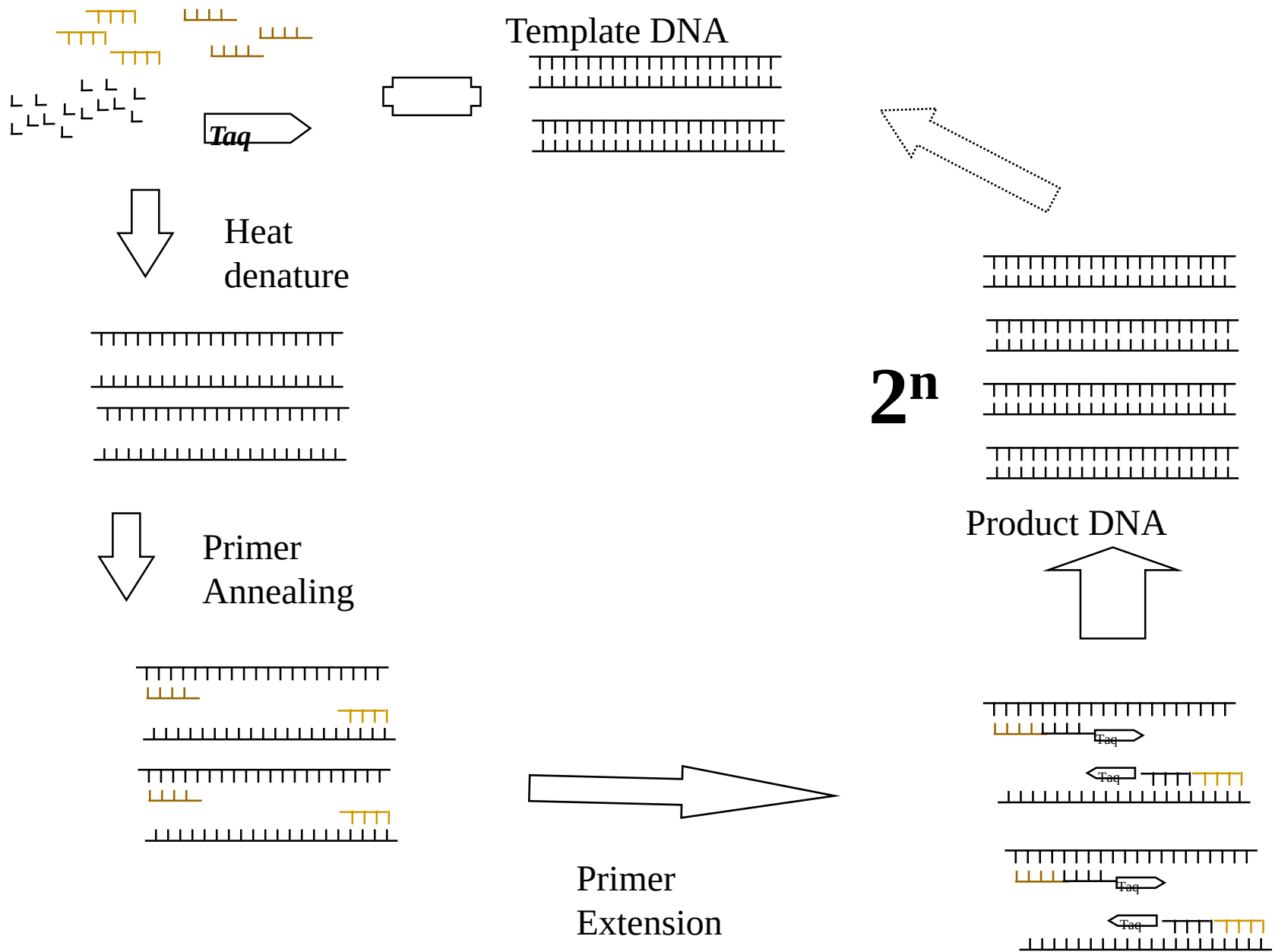
PCR (Step II)

■ Primer annealing

- Usually carried out 40-72°C
 - ✓ Determined by melting temperature of primer
 - ✓ Affected by primer/template mismatching
 - ✓ Affected by ionic composition of buffer, eg sodium / potassium ion conc.
- Rule of thumb for 15-25mer oligonucleotides:
 - ✓ for each G or C $T_m = 4^\circ\text{C}$
 - ✓ for each A or T $T_m = 2^\circ\text{C}$
 - ✓ Set PCR T_m at the estimated $T_m - 5^\circ\text{C}$
- For the following primer AGCTGATCGGTATCGGCT

PCR (Step III)

- Primer extension
 - At or near temperature of optimal enzyme activity
 - ✓ 68-72°C
 - ✓ Time interval determined by template/amplification target



PCR Reagents

■ Template DNA

- Range of DNA templates can be used which vary in purity, e.g. boiled bacterial colony to ultra-pure DNA sample
- Relative amount of target to be amplified important
 - ✓ Human gene has at least 2 copies per diploid cell
 - ✓ Bacterial plasmid 1-20 copies per cell
 - ✓ Virus (eg HIV) sometimes only 1 copy in 150,000 cells!

■ Primers

- Two primers required; one complementary to the sense and one complementary to the antisense strand
 - Sense primer
 - ✓ Upstream of region of interest (5' primer);
 - ✓ Anneals to antisense strand of target DNA
 - Antisense primer
 - ✓ Downstream of region of interest (3' primer);
 - ✓ Anneals to sense strand of target DNA
- Usually 15-40 nucleotides in length

- Optimum concentration determined by titration - typical conc. = 0.1-1.0 μM
- Chemically synthesized
- Annealing/melting temperature determined by base composition
 - A general rule if full complementarity between primer and template
 - ✓ A or T = 2°C and C or G = 4°C
- Can include modifications
 - Restriction enzyme sites at 5' end
 - Biotinylation or label incorporation

■ PRIMER DESIGN IS VITAL

- Length ~ 18-30nt (21nt)
- Base composition; 50 - 60% GC rich
pairs should have equivalent Tms
- $T_m =$
 $[(\text{number of A+T residues}) \times 2] + [(\text{number of G+C residues}) \times 4] ^\circ\text{C}$
- Initial use $T_m - 5^\circ\text{C}$
- Avoid internal hairpin structures
- Avoid a T at the 3' end
- Avoid overlapping 3' ends - will form primer dimers
- Can modify 5' ends to add restriction sites etc

✳ PRIMER DESIGN

Use specific programs

OLIGO
Medprobe

PRIMER DESIGNER
Sci Ed software

SECentral
File View Primer Operations Window Help

Search 1
Rank sort

Molecule: SYNPR322 4361 bps
Search for: PCR Primer Pair
Target Region: 86-1276;

Rank	Position		GC
	A	B	
1	67	1398	59
2	66	1360	54
3	69	1319	54
4	66	1319	54
5	67	1397	59
6	65	1397	54
7	65	1398	54
8	65	1360	54
9	65	1319	54

[Search 1] Primer Pair

Selected Pair:
Rank: 1 PCR Primer Pair Search
Primer A: Molecule: SYNPR322 Pos: 67
Primer B: Molecule: SYNPR322 Pos: 1398 Compl

A ☒ Length: 22 ACGCAGTCAGGCACCGTGTATG
B ☒ Length: 22 GGCGGACGCGATGGATATGTTT

Amplified product: Size: 1332 bps Product GC 60%
Calc temperatures: Melting T 84°C, Annealing T 60°C

Molecule: SYNPR322 Pos: 67, 1398 C

tet bla

Info Analyze

Also available on the internet

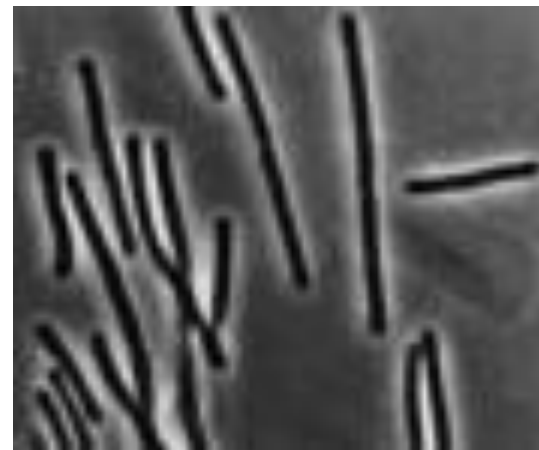
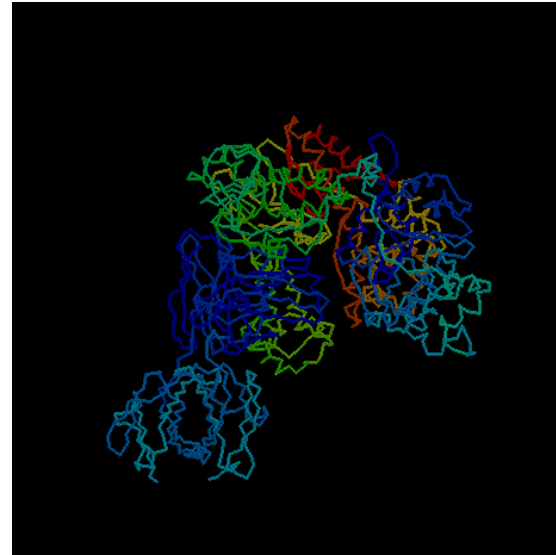
<http://www.hgmp.mrc.ac.uk/GenomeWeb/nuc-primer.html>

■ Buffer

- Buffer - provides correct chemical, ionic environment for enzyme function
 - pH at room temp high (pH9.0) this decreases with increasing temperature
 - Magnesium ion concentration determines enzyme activity
 - ✓ Optimum magnesium concentration template, primer and enzyme-specific, too low = no products and too high yields non-specific amplification products
 - ✓ Determined by titration
 - ✓ Typical concentration for most applications 1.5-2.5 mM final MgCl_2
 - Ionic strength affects primer annealing

Choice of Enzyme

- Originally used Klenow fragment
 - Thermolabile
 - Low temperature = low fidelity
- Thermostable enzymes (1988)
 - *Thermus aquaticus* (Taq)
 - Other thermostable enzymes
 - ✓ Pwo,
 - ✓ Pfu,
 - ✓ Taq mixes,
 - ✓ Tth (for RT activity)



Which enzyme to chose?

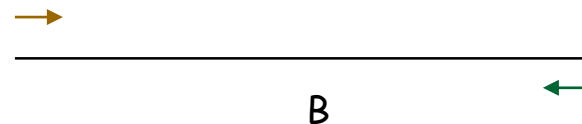
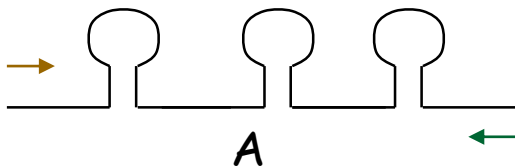
- Fidelity of enzyme
 - Taq = low, Pfu = high
- Processivity (진행도)
 - Rate of incorporation of nucleotides
 - Secondary structures cause enzyme pausing and even dissociation
 - Determines extension time - template specific

Other Reagents

- Nucleotides - DNA building blocks
 - Typically used at 200uM
- Mineral oil
 - Prevents evaporation
 - Not required in modern thermal cyclers
- Occasional additives
 - DMSO, TMAC (Trimellitic anhydride chloride),

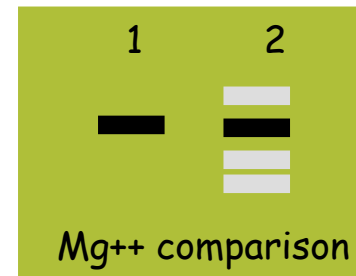
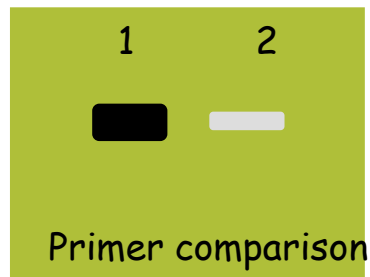
Other Considerations

- Primer mismatching
 - 3' end important - use of GC clamps
 - Affects melting temperature
 - Redundancy to overcome problem
- Secondary structures in template or primer
 - Predict using primer design programmes
 - ✓ Which of the following do you think will be easier to amplify:

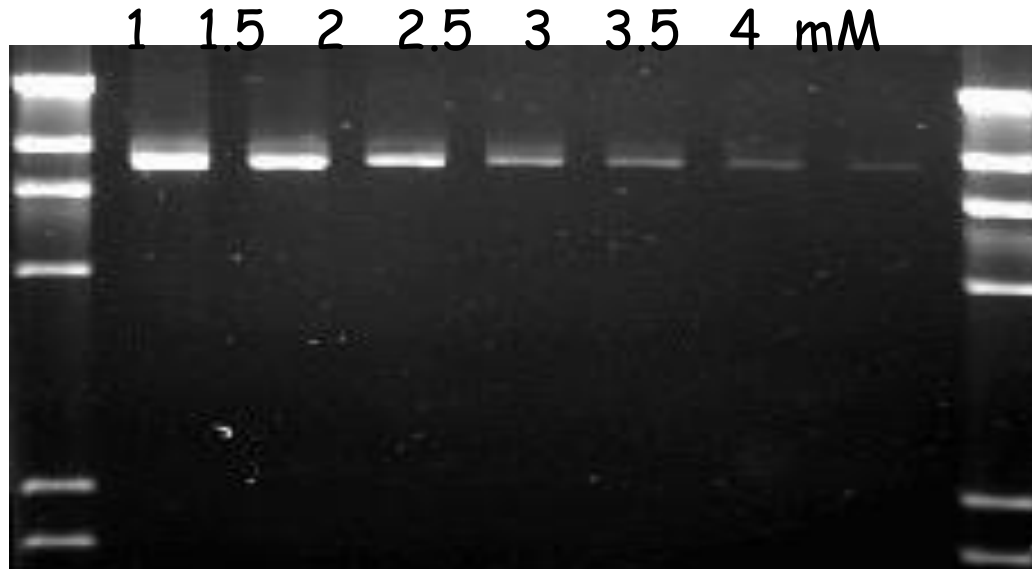


Optimization

- Optimization of a PCR is, on the whole, empirical
- General rule!
 - High stringency gives rise to more specific products
 - ✓ Increase annealing temperature
 - ✓ Decrease Mg^{++}
 - ✓ Decrease monovalent cations
 - ✓ Decrease primer concentration
 - Which of the following lanes represents products from high and low Mg^{++} concentration and high and low primer concentration?



※ TITRATE YOUR Mg^{2+} CONCENTRATION!



Normally, 1.5mM $MgCl_2$ is optimal

Best supplied as separate tube

Always vortex thawed $MgCl_2$

Mg^{2+} concentration will be affected by
the amount of DNA, primers and
nucleotides

Types of PCR

■ RT-PCR

- Used for RNA eg. RNA viruses or mRNA
 - ✓ Separate RT
 - AMV or MuLV RT - performed at 37- 42°C
 - Sequence-specific or random hexamer primer
 - Use RT products in PCR
 - ✓ One tube - often using thermostable RT
 - Tth has RT activity in Mn ions

■ Hot-Start PCR

■ Nested primer PCR

- ※ Both of these increase sensitivity and specificity

Types of PCR (Cont...)

- Discriminatory PCR
- In-situ PCR (IS-PCR)
 - cell - organism is the test-tube
- Multiplex PCR
 - multiple primer pairs
- RAPD
- Asymmetric PCR

PCR Uses

- Detection of pathogens
 - Qualitative
 - Quantitative
- First step in gene analysis
 - Sequencing templates
 - Cloning
Specially designed PCR cloning vectors
 - In vitro transcription
Protein study eg protein truncations
- Epidemiological(전염병학상의) studies

Summary

- PCR
 - Used to amplify nucleic acids (DNA)
RNA needs first RT step
 - Highly sensitive
 - First stage in many molecular biology applications
- Variety of methods
 - nested primer, multiplex, RAPD etc. etc.

2. 유전자의 대장균 내 도입방법

➤ 형질전환 (Transformation)

- 외부 DNA를 숙주세포에 넣어 세포가 원래의 성질과는 다른 새로운 유전형질을 추가로 얻게 되는 것
- 1928년 그리피스: 비병원성 폐렴쌍구균 → 병원성 균
- 항생제 저항성 유전자를 함유한 플라스미드 벡터를 항생제에 민감한 박테리아에 도입

➤ 형질전환의 이용

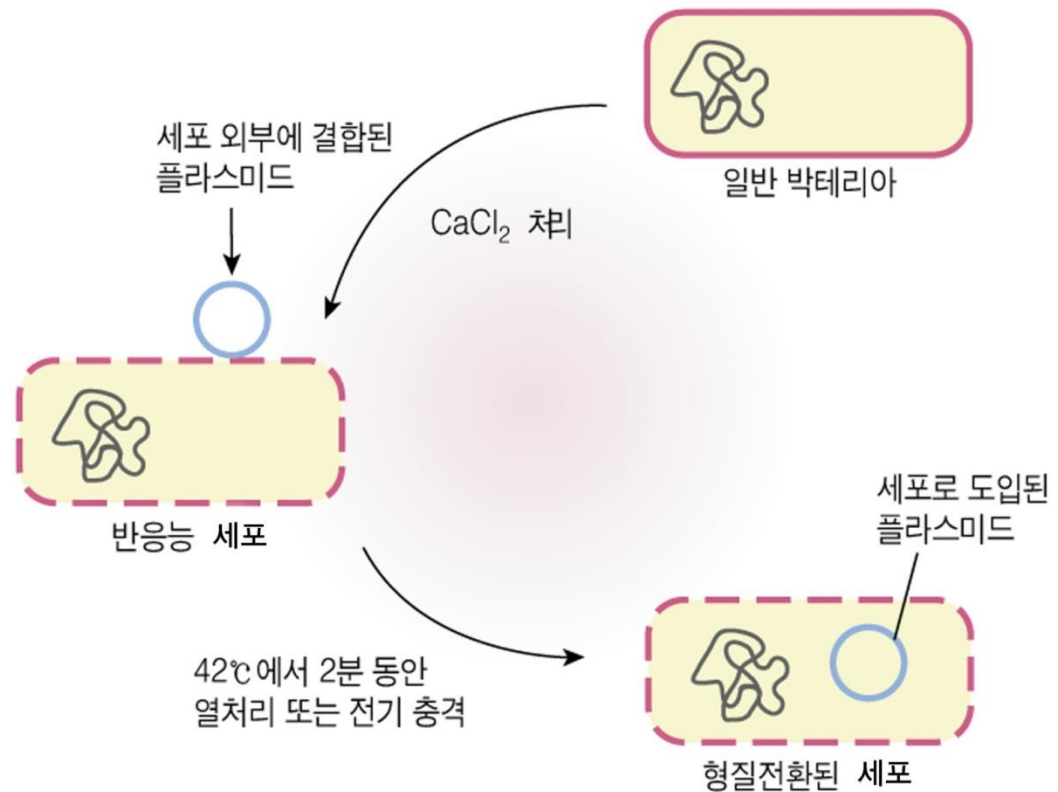
- 단백질 산물 획득
- 기능성 생명체 획득
- 유전자 클로닝 기능: 다수의 재조합 DNA 분자 획득

➤ 대장균 균주

- *Escherichia coli* (*E. coli*)
- 포유류의 대장에서 서식하는 박테리아의 대표 종
- 박테리아의 모델 생물
- 길이: 2 μm , 폭: 0.8 μm
- 약 400 만개의 염기쌍, 약 3000개의 유전자
- 하루에 약 72번 분열
- 실험실에서 흔히 사용되는 대장균 균주
- DH5 α , XL1-Blue, BL21, JM109

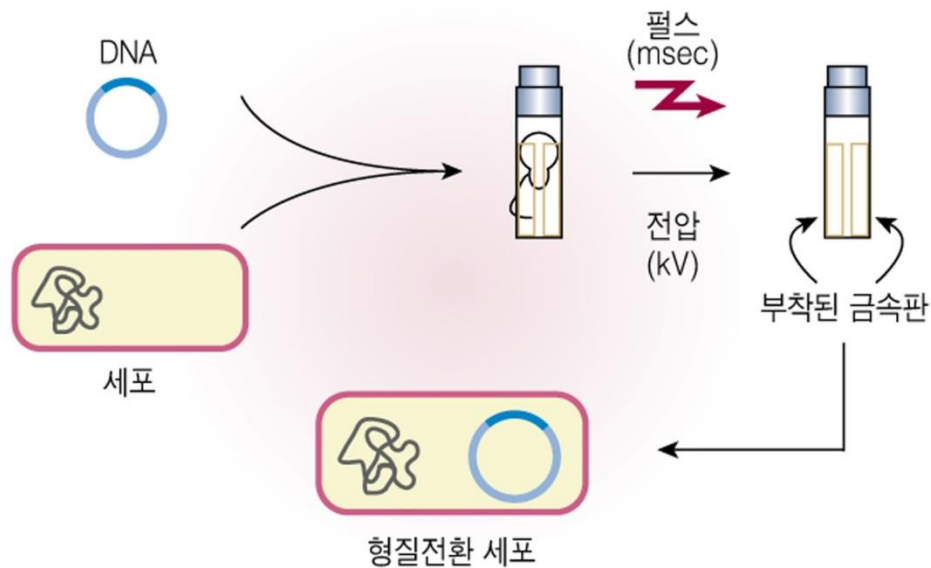
➤ 반응능 세포 (competent cell)

- 대장균이 외부 DNA를 받아 들일 수 있는 상태로 만들어 준 세포
- CaCl_2 용액을 이용

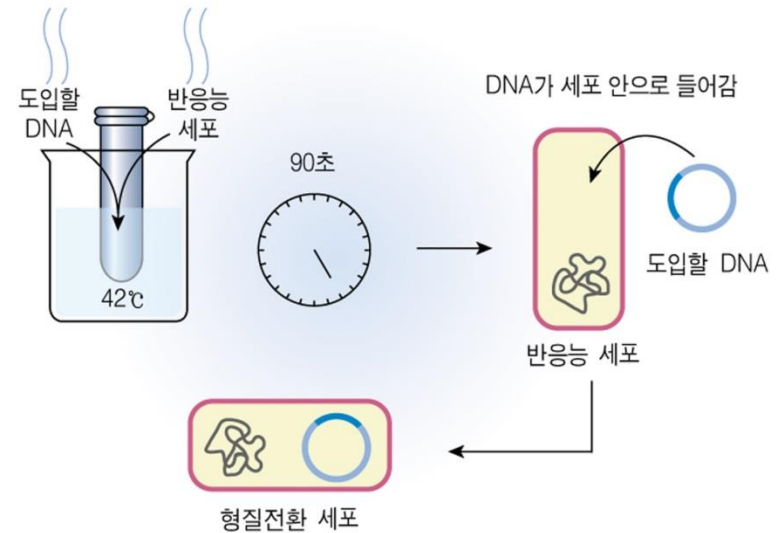


➤ 반응능 세포로의 DNA 도입 방법

- Heat shock
- Electroporation



반응능 세포와 도입할 DNA를
함께 열 충격을 줌



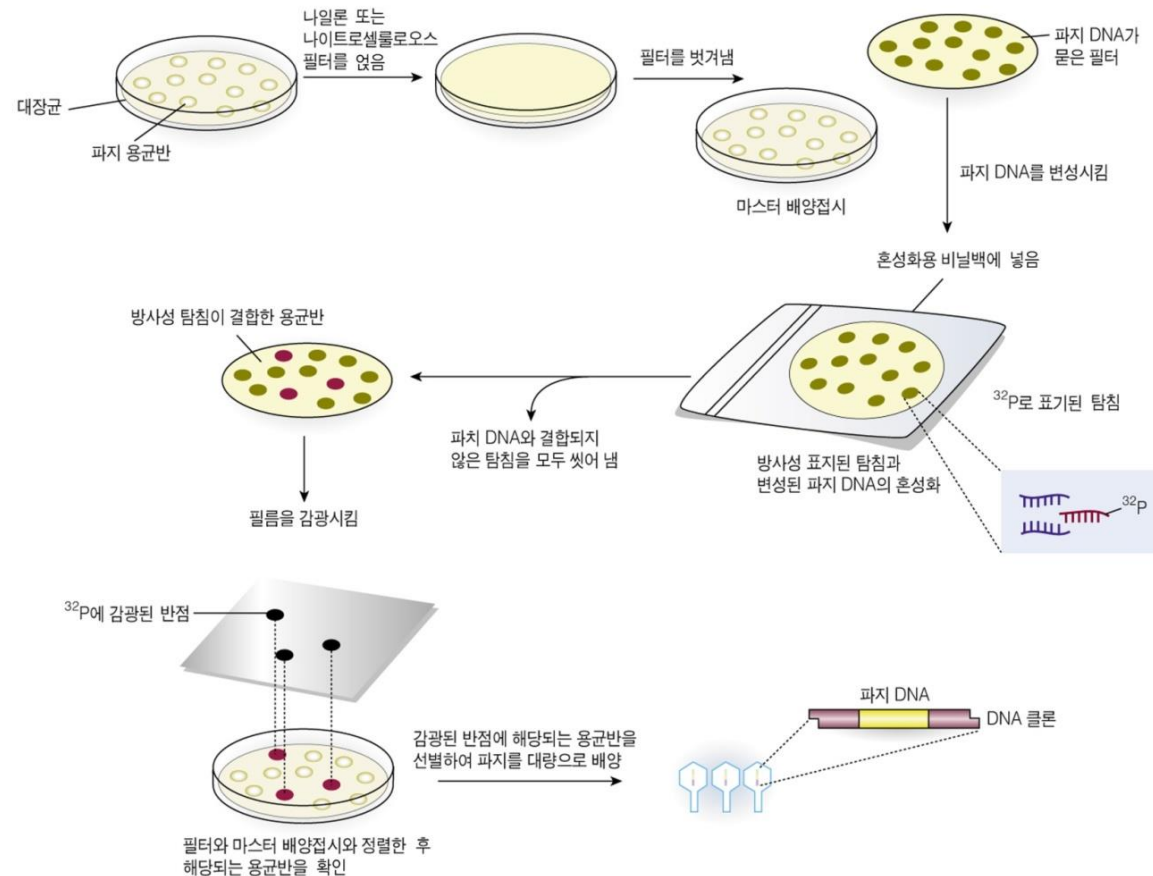
3. 형질전환된 세포의 선발/ Screening

- 라이브러리속에 존재하는 표적 유전자를 골라내는 작업
 - DNA 혼성화를 이용하는 방법
 - 특정 단백질 항체를 이용하는 방법
 - 유전자 발현 차이를 이용하는 방법
 - 특정 단백질과의 결합을 이용하는 방법
- 혼성화를 이용하는 방법
 - 한가닥으로 된 핵산이 이와 상보적인 염기서열의 또 다른 가닥의 핵산과 이중나선을 형성하는 현상
 - ✓ Stringency (긴축도):
 - High stringency
 - Low stringency
 - ✓ Probes (탐침)

■ Probes (탐침)

✓ 방사선 표지 (labeling)

- ❑ ^{32}P labeled dCTP or dATP로 DNA 합성
- ❑ Autoradiography (방사선자기사진법)



➤ 재조합 플라스미드 벡터의 확인

- 항생제에 의한 선발
- **Antibiotics**
- 중요 항생제의 종류와 작용
 - ✓ 세포벽 합성 저해제
 - ☐ Penicillin (β -lactam)
 - ☐ Cephalosporin
 - ✓ 테트라하이드로 엽산(THF: Tetrahydrofolate) 합성 저해제
 - ☐ THF: 퓨린염기와 티민 합성에 필요한 조효소
 - ☐ Sulfonamide
 - ✓ DNA 복제 저해제
 - ☐ Gyrase 저해제: ofloxacin (quinolone 유도체)
 - ✓ 단백질 합성 저해제
 - ☐ Tetracycline
 - ☐ Kanamycine

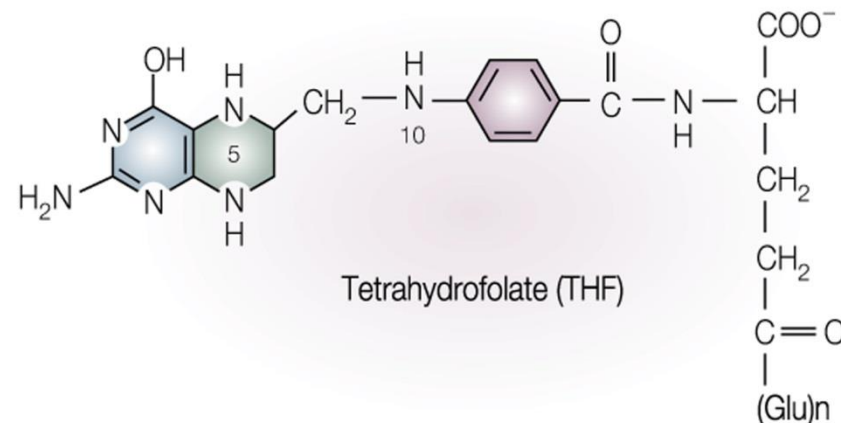


표 6.1 분자생물학 실험에 주로 이용되는 중요 항생제의 종류

항생제 구분	종 류
세포벽 합성 저해제	페니실린(penicillin) A. 자연 페니실린(natural penicillin): Penicillin G, Procaine Penicillin G, Penicillin V, Benzathine B. 아미노페니실린(aminopenicillin): 앰피실린(Ampicillin), Amoxicillin, Augmentin, Unasyn C. Extended Spectrum penicillin: Ticarcillin, Piperacillin, Carbenicillin, Timentin) D. Penicillin-Resistant Semisynthetic Penicillin: Cloxacillin, Dicloxacillin, Methicillin, Nafcillin, Oxacillin 세파로스포린(cephalosporin), 바시트라신(bacitracin), 반코마이신(vancomycin)
테트라하이드로엽산 (THF) 합성 저해제	설파제(sulfonamide)류와 트리메토프림(trimethoprim)
DNA 복제 저해제	오픈록사신(Ofloxacin) 등과 같은 퀴놀론 유도체
단백질 합성 저해제	카나마이신(kanamycin), 테트라사이클린(tetracycline), 아미노글리코사이드(aminoglycoside), 클로람페니콜(chloramphenicol), 에리트로마이신(erythromycin)

➤ 항생제 선발 표지자

- Ampicillin에 저항성을 나타내는 β -lactamase 유전자
- Kanamycin에 저항성을 나타내는 neomycin phosphotransferase type II 유전자

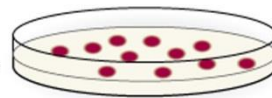
앰피실린 처리 시

항생제 저항 유전자를 포함
하는 플라스미드가 없음



콜로니 형성 불가

항생제 저항 유전자를 포함
하는 플라스미드가 있고
재조합 유전자도 포함함



콜로니 형성 가능

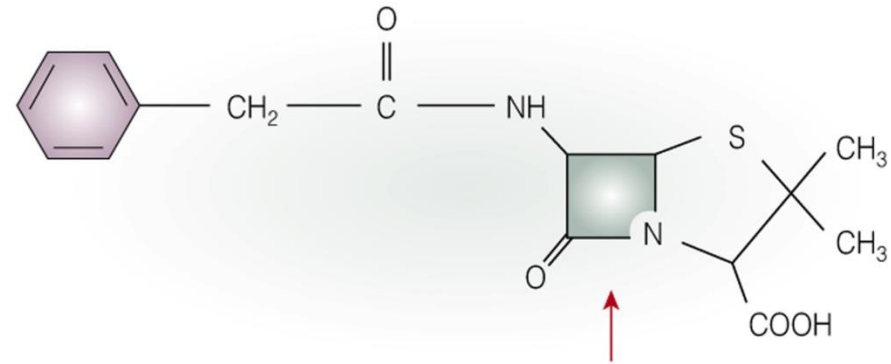


표 6.2 분자생물학 실험에서 많이 사용되는 항생제 선발 표지자

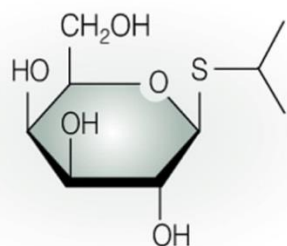
항생제	항생제 작용 메커니즘	선발 표지 유전자 및 작용
엠포실린(ampicillin)	펩티도글리칸 결합을 방해하여 세포벽 합성 억제	β -락타메이즈(β -lactamase)가 엠포실린을 분해
클로람페니콜(chloramphenicol)	50S 리보솜의 펩티딜전달효소의 작용을 방해하여 단백질 합성 억제.	chloramphenicol acetyltransferase가 클로람페니콜의 작용을 저해
젠타마이신(gentamycin)	50S 리보솜에 결합하여 단백질 합성 억제	aminoglycoside acetyltransferase와 aminoglycoside nucleotidyltransferase가 젠타마이신 작용을 저해
하이그로마이신(hygromycin)	단백질 합성 억제	Hygromycin phosphotransferase가 hygromycin의 작용을 저해
카나마이신(kanamycin)	70S 리보솜에 결합하여 단백질 합성 억제	neomycin phosphotransferase가 카나마이신의 작용을 저해
리팜피신(rifampicin)	RNA 중합효소의 작용을 방해하여 전사 억제	<i>rpoB</i> 유전자 산물이 리팜피신이 RNA 중합효소에 결합하지 못하게 함.
스펙티노마이신(spectinomycin)	전좌를 방해하여 단백질 합성 억제	5S 리보솜 단백질을 암호화하는 <i>rpsE</i> 유전자의 아미노산 치환 돌연변이에 의해 스펙티노마이신이 결합하지 못함
스트렙토마이신(streptomycin)	370S 리보솜에 결합하여 단백질 합성 억제	아미노글리코사이드 인산기전달효소(aminoglycoside phosphotransferase)가 스트렙토마이신을 저해
테트라사이클린(tetracycline)	아미노아실 tRNA의 리보솜 결합을 방해하여 단백질 합성 억제	테트라사이클린 억제 단백질 생성 유전자에 의한 세포의 방출(efflux) 조절

3-2. *LacZ* (β -galactosidase)의 삽입 비활성화에 의한 선발

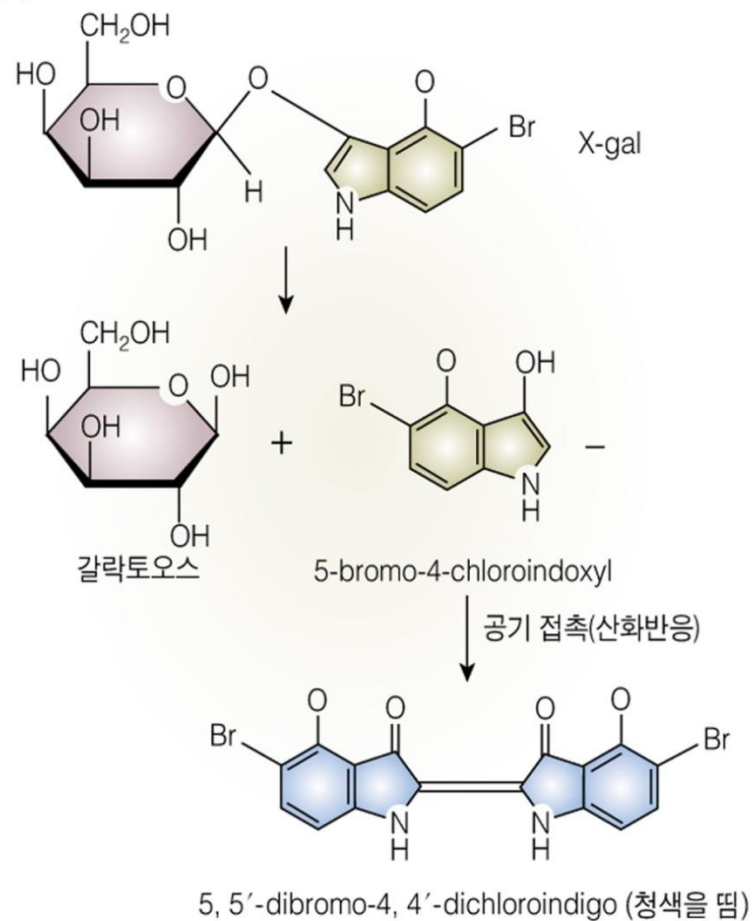
➤ α -complementation

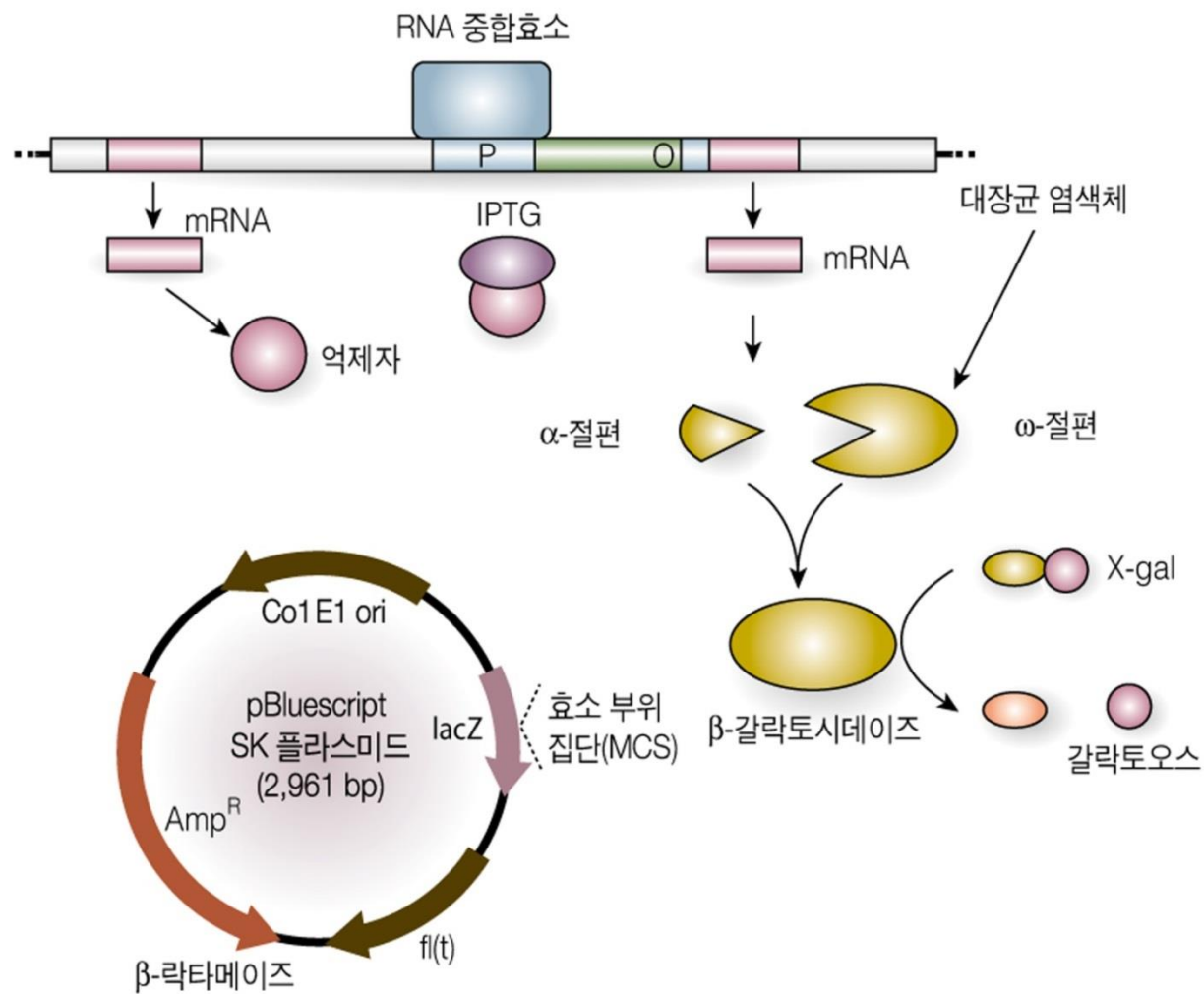
- 플라스미드의 *lacZ* 유전자:
 - ✓ β -galactosidase의 일부분인 α -절편을 생성
 - 대장균 *lacZ* 유전자
 - ✓ β -galactosidase의 ω -절편을 생성
 - 두 부분이 합쳐져 완전한 효소를 이룸
- 플라스미드의 ***lacZ*** 유전자는 젓당 유사물질인 IPTG에 의해 발현이 유도 됨
- X-Gal (5-Bromo-4-chloro-3-indolyl- β -D-galactoside) 시약을 사용 β -galactosidase 유전자의 손상을 확인하여 목적 DNA 삽입 여부를 확인 함

(A)

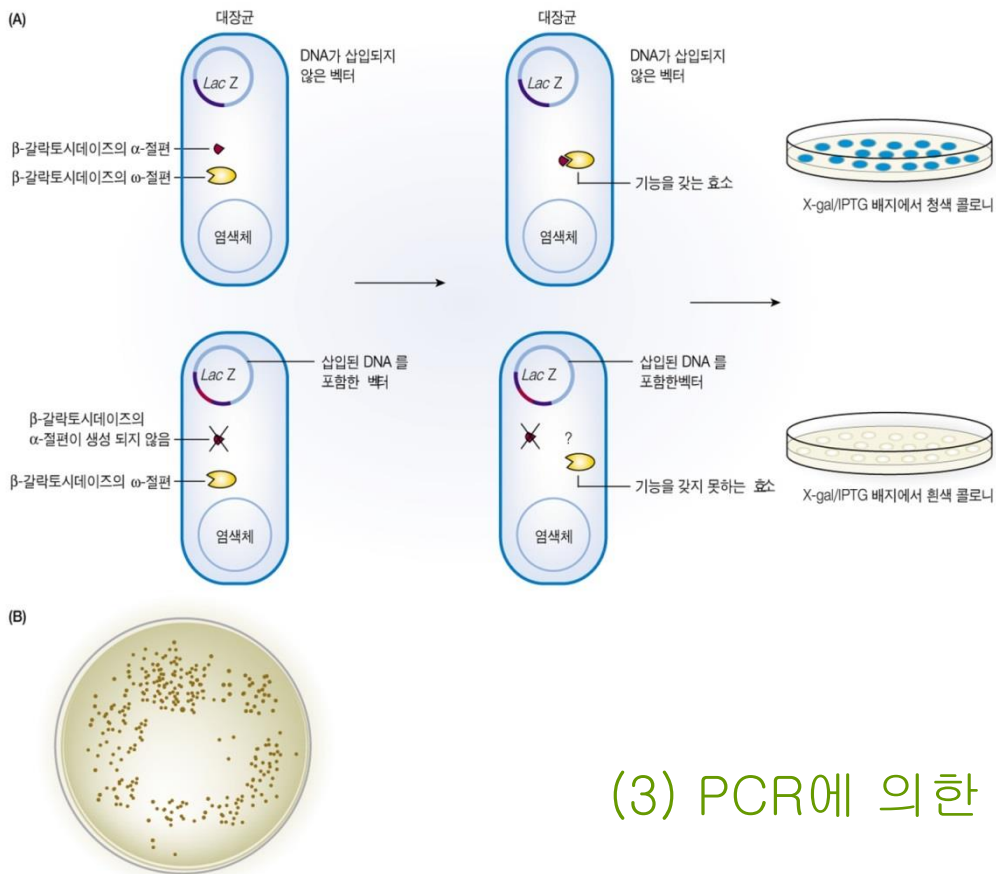
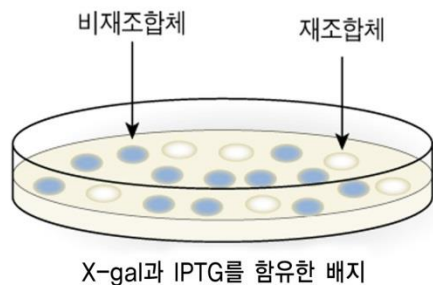


(B)



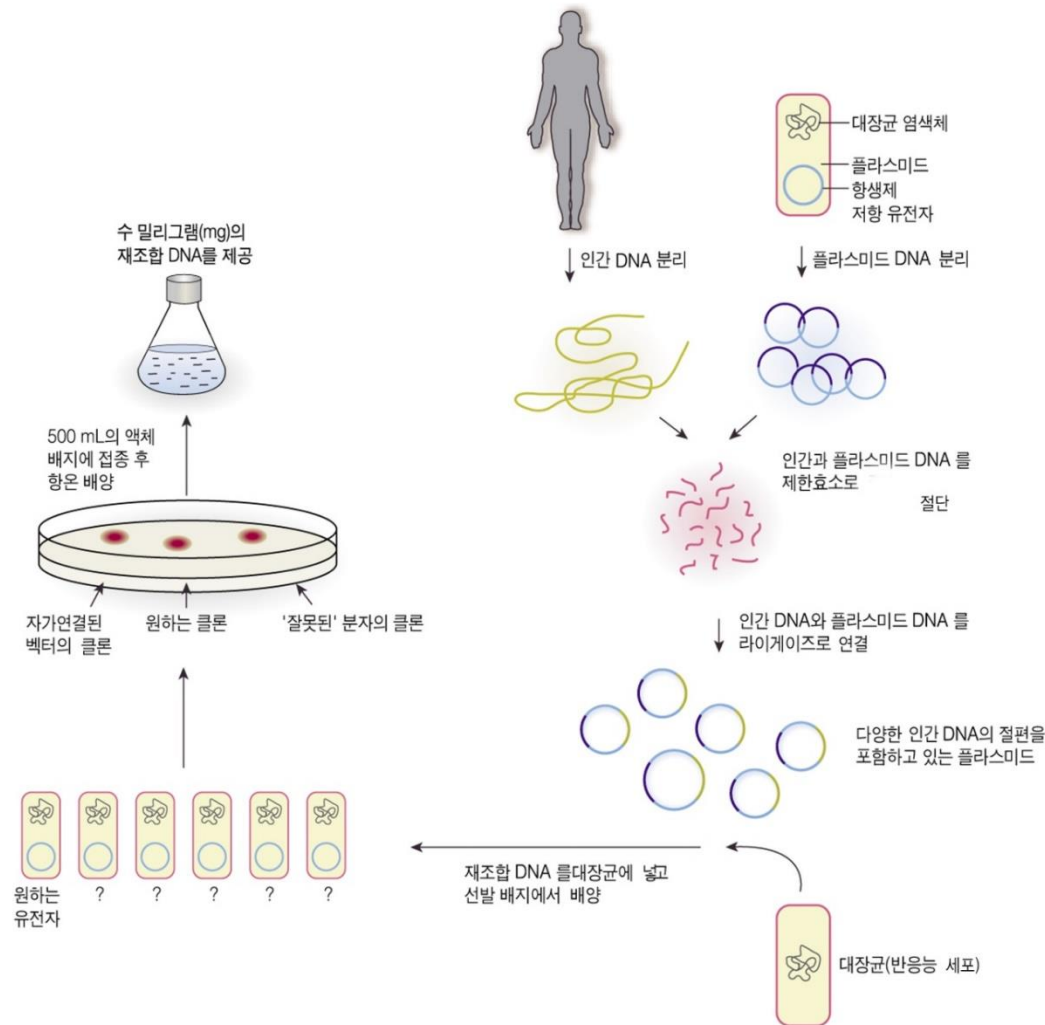


- 대장균은 본래 β -galactosidase를 생성할 수 있는 유전자를 가짐
- α -complementation에 사용되는 균주는 lacZ의 ω -절편 유전자가 제거된 돌연변이 균주 (DH5 α , JM109 Δ M15)



(3) PCR에 의한 선발

4. DNA 재조합 기술의 전과정

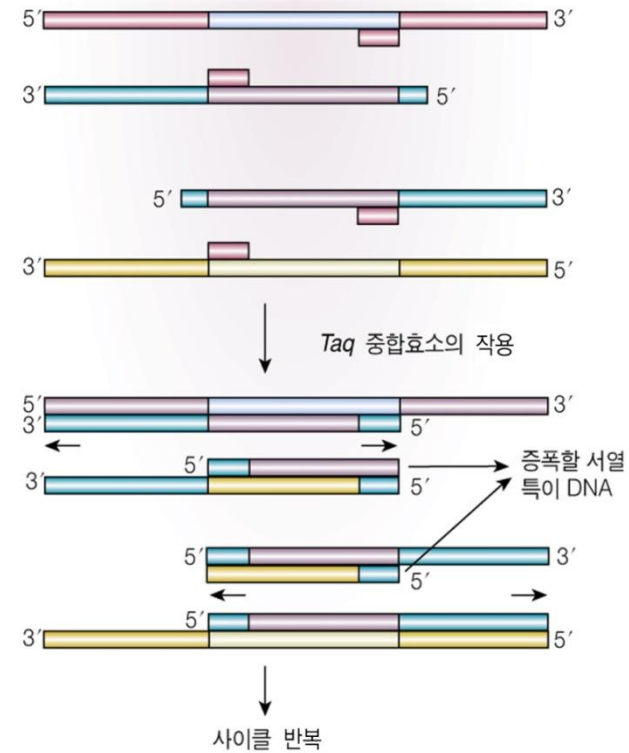
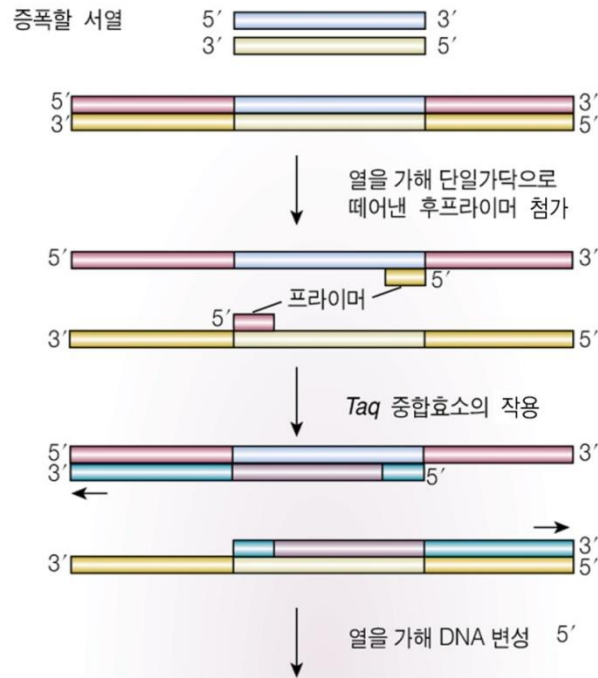




➤ Getting the target DNA

1. Gene Libraries
2. PCR
3. Chemical synthesis

1. PCR



5'...CTGACACAACCTGTGTTCACTAGCAA AAGGTGAACGTGGATGAAGTTGGTG...3'
 3'...GACTGTGTTGACACAAGTGATCGTT TTCCACTTGCACCTACTTCAACCAC...5'

↓ 가열

5'...CTGACACAACCTGTGTTCACTAGCAA AAGGTGAACGTGGATGAAGTTGGTG...3'
 3'...GACTGTGTTGACACAAGTGATCGTT TTCCACTTGCACCTACTTCAACCAC...5'

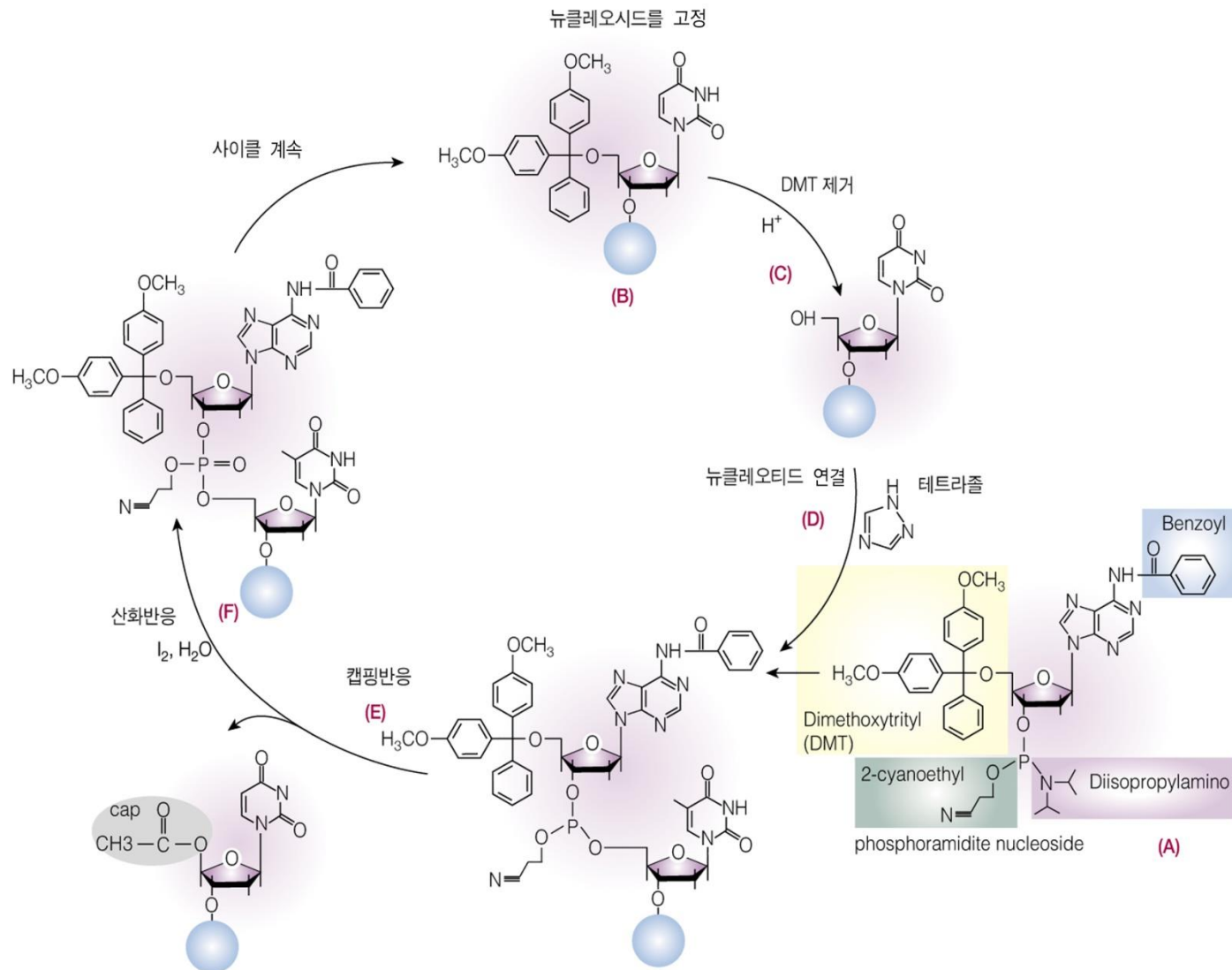
↓ 프라이머 결합

5'...CTGACACAACCTGTGTTCACTAGCAA AAGGTGAACGTGGATGAAGTTGGTG...3'
 정방향 프라이머 3' CCACCTTGCACCTACTTCAAC 5'
 5' ACACAACCTGTGTTCACTAGC 3' 역방향 프라이머
 3'...GACTGTGTTGACACAAGTGATCGTT TTCCACTTGCACCTACTTCAACCAC...5'

↓ 중합효소 작용 / 사슬 신장

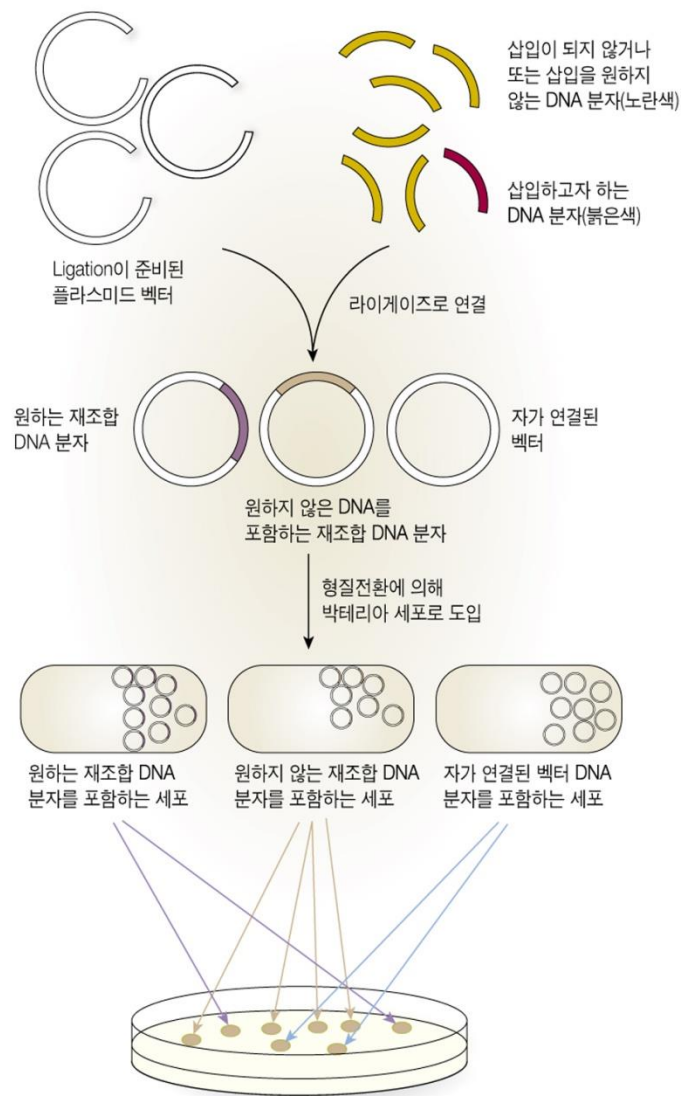
5'...CTGACACAACCTGTGTTCACTAGCAA AAGGTGAACGTGGATGAAGTTGGTG...3'
 3'...TTCCACTTGCACCTACTTCAAC 5'
 5' ACACAACCTGTGTTCACTAGCAA...3' →
 3'...GACTGTGTTGACACAAGTGATCGTT TTCCACTTGCACCTACTTCAACCAC...5'

4. 화학적 DNA 합성



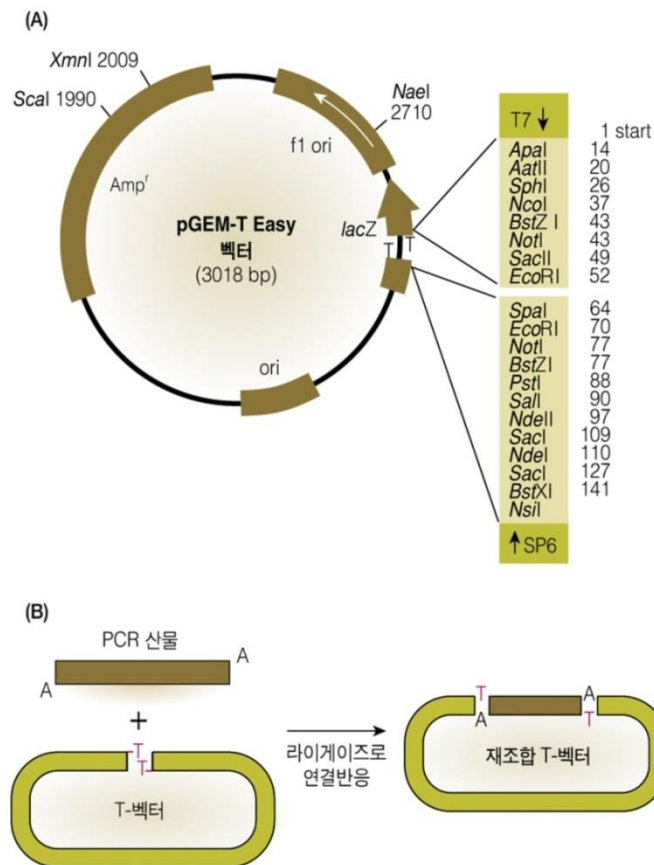
➤ Cloning

- 두종류의 제한효소처리가 가능한 경우 자가연결방지 및 삽입 원하는 방향으로 진행



➤ PCR 산물을 이용한 재조합 DNA 분자의 제작

- Primer design: next cloning 작업을 고려
- TA cloning
- 프라이머에 제한효소자리를 이용한 클로닝



- 형질 전환 (Transformation)
- Screening (Blue-White screening)